

A new golden age of discovery

发现的新黄金时代

Seizing the AI for Science opportunity

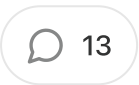
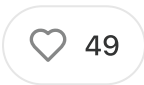
把握 AI 为科学带来的机遇



AI POLICY PERSPECTIVES, CONOR GRIFFIN, HANNA SCHIEVE, AND 2 OTHERS

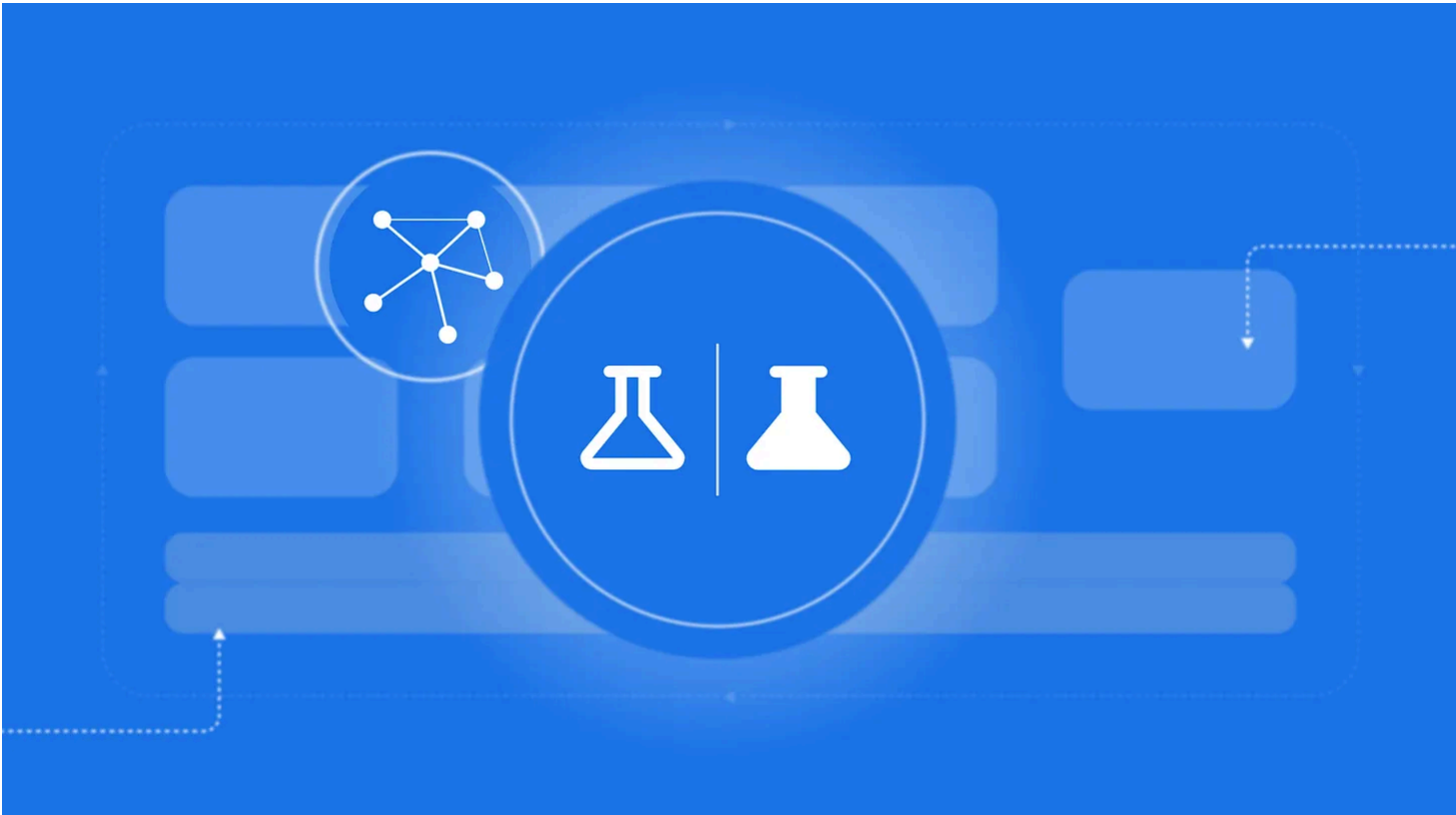
AI 政策视角，康纳·格里芬，汉娜·希夫，以及其他 2 人

NOV 26, 2024 2024 年 11 月 26 日



This essay is written by [Conor Griffin](#), [Don Wallace](#), [Juan Mateos-Garcia](#), [Hanna Schieve](#) and [Pushmeet Kohli](#). Subscribe to receive future essays and leave a comment below to give us your take on the most important AI for Science opportunities, ingredients, risks and public policy ideas. This essay was originally posted [on the Google DeepMind website](#).

这篇文章由康纳·格里芬、唐·华莱士、胡安·马特奥斯-加西亚、汉娜·希夫和普什梅特·科利撰写。订阅以获取未来的文章，并在下方留言，分享您对最重要的科学 AI 机遇、要素、风险和公共政策想法的看法。这篇文章最初发布在 Google DeepMind 网站上。



Introduction 引言

A quiet revolution is brewing in labs around the world, where scientists' use of AI is [growing exponentially](#). [One in three postdocs](#) now use large language models to help carry out literature reviews, coding, and editing. In October, the creators of our [AlphaFold 2](#) system, Demis Hassabis and John Jumper, became [Nobel](#) Laureates in Chemistry for using AI to predict the structure of proteins, alongside the scientist David Baker, for his work to design new proteins. Society will soon start to feel these benefits more directly, with [drugs](#) and materials designed with the help of AI currently making their way through development.

一场安静的革命正在全球实验室中酝酿，科学家们对 AI 的使用正在呈指数级增长。现在，每三位博士后中就有一位使用大型语言模型来帮助进行文献综述、编码和编辑。今年 10 月，我们的 AlphaFold 2 系统的创造者德米斯·哈萨比和约翰·琼珀，以及科学家大卫·贝克因使用 AI 预测蛋白质结构（贝克则因设计新蛋白质的工作）而共同获得诺贝尔化学奖。社会很快将更直接地感受到这些益处，目前有 AI 辅助设计的药物和材料正在开发过程中。

In this essay, we take a tour of how AI is transforming scientific disciplines from genomics to computer science to weather forecasting. Some scientists are training their own AI models, while others are fine-tuning existing AI models, or using these models' predictions to accelerate their research. These scientists are using AI as a scientific instrument to help tackle important problems, such as [designing proteins that bind more tightly to disease targets](#), but are also gradually transforming how science itself is practised.

在这篇论文中，我们探讨了人工智能如何从基因组学到计算机科学再到天气预报等领域改变着科学学科。一些科学家正在训练自己的 AI 模型，而另一些科学家则在微调现有的 AI 模型，或利用这些模型的预测来加速他们的研究。这些科学家正将 AI 作为科学工具来帮助解决重要问题，例如设计能更紧密结合疾病靶点的蛋白质，同时也逐渐改变着科学本身的实践方式。

There is a growing imperative behind scientists' embrace of AI. In recent decades, scientists have continued to deliver consequential advances, from Covid-19 vaccines to renewable energy. But it takes [an ever larger number of researchers to make these breakthroughs](#), and to [transform them into downstream applications](#). As a result, even though the scientific workforce has grown significantly over the past half-century, [rising more than seven fold](#) in the US alone, the societal progress that we would

expect to follow, has slowed. For instance, much of the world has witnessed a [sustained slowdown](#) in productivity growth that is undermining the quality of public services. Progress towards the 2030 Sustainable Development Goals, which capture the biggest challenges in health, the environment, and beyond, is [stalling](#).

科学家们拥抱人工智能背后，存在着日益增长的紧迫感。在近几十年里，科学家们持续取得了重大进展，从新冠疫苗到可再生能源。但是，要取得这些突破并将其转化为下游应用，需要越来越多的研究人员。因此，尽管过去半个世纪以来，科学工作队伍显著增长，在美国 alone 就增长了七倍以上，但社会进步却放缓了。例如，世界上许多国家都目睹了生产率增长的持续放缓，这正在削弱公共服务的质量。朝着 2030 年可持续发展目标（这些目标涵盖了健康、环境等最大挑战）的进展正在停滞不前。

In particular, scientists looking to make breakthroughs today increasingly run into challenges relating to *scale* and *complexity*, from the ever-growing literature base they need to master, to the increasingly complex experiments they want to run. [Modern deep learning methods](#) are particularly well-suited to these [scale and complexity challenges](#) and can compress the time that future scientific progress would otherwise require. For instance, in structural biology, a single x-ray crystallography experiment to determine the structure of a protein [can take years of work and cost approximately \\$100,000](#), depending on the protein. The [AlphaFold Protein Structure Database](#) now provides instant access to 200 million predicted protein structures for free.

特别是，今天希望取得突破的科学家们越来越多地遇到规模和复杂性的挑战，从他们需要掌握的日益增长的文献基础，到他们想要进行的日益复杂的实验。现代深度学习方法特别适合这些规模和复杂性的挑战，并且可以压缩未来科学进步所需的时间。例如，在结构生物学中，一个用于确定蛋白质结构的 X 射线晶体学实验可能需要数年的工作，并且成本大约为 10 万美元，具体取决于蛋白质。AlphaFold 蛋白质结构数据库现在免费提供对 2 亿个预测的蛋白质结构的即时访问。

The potential benefits of AI to science are not guaranteed. A significant share of scientists already use LLM-based tools to assist with everyday tasks, such as coding and editing, but the share of scientists using AI-centric research approaches [is much lower, albeit rising rapidly](#). In the rush to use AI, some early scientific use cases have had [questionable impact](#). Policymakers can help accelerate AI's use and steer it towards higher-impact areas. The [US Department of Energy](#), the [European Commission](#), the UK's [Royal Society](#), and the [US National Academies](#), among others, have recently recognised the *AI for Science* opportunity. But no country has yet put a

comprehensive strategy in place to enable it.

人工智能对科学的潜在益处并非必然。已有相当一部分科学家使用基于 LLM 的工具来协助日常任务，如编程和编辑，但采用以 AI 为中心的研究方法的科学家比例要低得多，尽管这一比例正在迅速增长。在急于使用 AI 的过程中，一些早期的科学应用案例产生了可疑的影响。政策制定者可以帮助加速 AI 的使用，并将其引导至高影响力领域。美国能源部、欧洲委员会、英国皇家学会以及美国国家科学院等机构，近期已认识到 AI for Science 的机遇。但至今还没有国家制定出全面的战略来支持这一机遇。

We hope our essay can inform such a strategy. It is aimed at those who make and influence science policy, and funding decisions. We first identify 5 opportunities where there is a growing imperative to use AI in science and examine the primary ingredients needed to make breakthroughs in these areas. We then explore the most commonly-cited risks from using AI in science, such as to scientific creativity and reliability, and argue that AI can ultimately be net beneficial in each area. We conclude with four public policy ideas to help usher in a new golden age of AI-enabled science.

我们希望我们的文章能够为制定此类策略提供信息。它面向那些制定和影响科学政策以及资金决策的人。我们首先确定了 5 个使用 AI 进行科学研究的迫切需求日益增长的领域，并考察了在这些领域取得突破所需的主要要素。然后，我们探讨了在科学研究中使用 AI 时最常被提及的风险，例如对科学创造性和可靠性的影响，并论证 AI 最终可以在每个领域带来净收益。最后，我们提出了四个公共政策建议，以帮助开启 AI 赋能科学的新黄金时代。

Throughout the essay we draw on insights from over two dozen interviews with experts from our own AI for Science projects at Google DeepMind, as well as external experts. The essay naturally reflects our vantage point as a private sector lab, but we believe the case we make is relevant for the whole of science. We hope that readers will respond by sharing their take on the most important AI for Science opportunities, ingredients, risks and policy ideas.

在整个文章中，我们借鉴了来自我们自己在谷歌 DeepMind 的 AI for Science 项目以及外部专家的二十多次访谈的见解。文章自然地反映了我们作为私营部门实验室的立场，但我们相信我们提出的案例对整个科学领域都具有重要意义。我们希望读者能够通过分享他们对 AI for Science 最重要的机遇、要素、风险和政策建议的看法来做出回应。

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Part A. The opportunities

第一部分. 机遇

Scientists aim to understand, predict, and influence how the natural and social worlds work, to inspire and satisfy curiosity, and to tackle important problems facing society. [Technologies and methods](#), like the microscope, x-ray diffraction, and statistics, are both *products* of science and *enablers* of it. Over the past century, scientists have increasingly relied on these instruments to carry out their experiments and advance their theories. Computational tools and large-scale data analysis have become particularly important, enabling everything from the discovery of the Higgs boson to the mapping of the human genome. From one view, scientists' growing use of AI is a logical extension of this long-running trend. But it may also signal something much more profound - a discontinuous leap in the limits of what science is capable of.

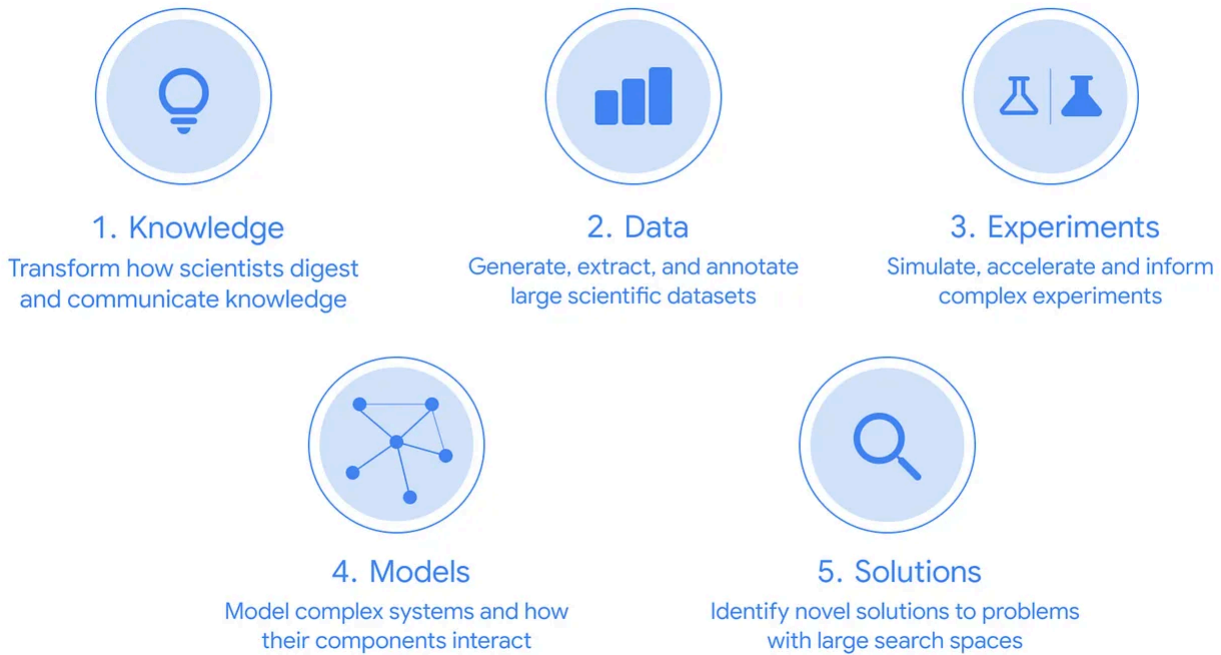
科学家致力于理解、预测和影响自然与社会世界的运作方式，激发并满足好奇心，以及解决社会面临的重要问题。显微镜、X 射线衍射和统计学等技术与方法，既是科学的产物，也是科学发展的推动力。在过去的一个世纪里，科学家越来越依赖这些仪器进行实验和推进理论。计算工具和大规模数据分析变得尤为重要，它们使得从发现希格斯玻色子到绘制人类基因组图谱等一切成为可能。从某种角度来看，科学家日益广泛地使用人工智能是这一长期趋势的合乎逻辑的延伸。但它也可能预示着更深远的变化——科学能力极限的跃迁式突破。

Rather than listing all areas where it is *possible* to use AI, we highlight five opportunities where we think there is an *imperative* to use it. These opportunities apply across disciplines and address a specific bottleneck, related to scale and complexity, that scientists increasingly face at different points in the scientific process,

from generating powerful novel hypotheses to sharing their work with the world.

与其列出所有可以使用 AI 的领域，我们着重强调五个我们认为有必要使用 AI 的机会。这些机会适用于各个学科，并解决一个特定的瓶颈问题，即规模和复杂性，这是科学家在科学过程的各个阶段（从生成强大的新假设到与世界分享他们的工作）越来越多地面临的挑战。

5 Opportunities to Accelerate Science with AI



1. KNOWLEDGE: Transform how scientists digest and communicate knowledge

1. 知识：改变科学家如何消化和交流知识

To make new discoveries, scientists need to master a pre-existing body of knowledge that continues to grow **exponentially** and become ever more specialised. This '**burden of knowledge**' helps explain why scientists **making transformative discoveries** are increasingly older, interdisciplinary, and located at elite universities, and why the **share of papers** authored by individuals, or small teams, is declining, even though **small teams are often better-placed to advance disruptive scientific ideas**. When it comes to sharing their research there have been welcome innovations such as preprint servers and code repositories, but most scientists still share their findings in **dense, jargon-heavy, English-only papers**. This can impede rather than ignite interest in scientists'

work, including from policymakers, businesses, and the public.

要做出新的发现，科学家需要掌握一个不断呈指数级增长且日益专业化的既有知识体系。这种“知识负担”有助于解释为何做出变革性发现的科学家越来越年长、跨学科，且多位于顶尖大学，也解释了为何由个人或小团队撰写的论文比例正在下降，尽管小团队往往更有能力推进颠覆性科学思想。在分享研究成果方面，虽然出现了预印本服务器和代码库等令人欢迎的创新，但大多数科学家仍通过密集、术语繁多且仅用英语撰写的论文来分享他们的发现。这可能会阻碍而非激发政策制定者、企业和公众对科学家工作的兴趣。

Scientists are already using LLMs, and early scientific assistants based on LLMs, to help address these challenges, such as by [synthesising the most relevant insights](#) from the literature. In an early [demonstration](#), our Science team used our [Gemini LLM](#) to find, extract, and populate specific data from the most relevant subset of 200,000 papers, within a day. Upcoming innovations, such as fine-tuning LLMs on more scientific data and advances in long context windows and citation use, will steadily improve these capabilities. As we expand on below, these opportunities are not without risk. But they provide a window to fundamentally rethink certain scientific tasks, such as what it means to ‘read’ or ‘write’ a scientific paper in a world where a scientist can use an LLM to help critique it, tailor its implications for different audiences, or transform it into an ‘interactive paper’ or [audio guide](#).

科学家们已经使用 LLMs 和基于 LLMs 的早期科学助手来帮助应对这些挑战，例如从文献中综合最相关的见解。在早期的一次演示中，我们的科学团队使用 Gemini LLM 在一天内从最相关的 20 万篇论文子集中找到、提取并填充特定数据。即将到来的创新，如使用更多科学数据微调 LLMs 以及长上下文窗口和引用使用的进步，将稳步提升这些能力。正如我们下面将扩展的，这些机会并非没有风险。但它们提供了一个窗口，让我们能够从根本上重新思考某些科学任务，例如在一个科学家可以使用 LLM 来帮助评论、为不同受众调整其影响或将之转化为“交互式论文”或音频指南的世界中，“阅读”或“撰写”科学论文意味着什么。

2. DATA: Generate, extract, and annotate large scientific datasets

2. 数据：生成、提取和标注大型科学数据集

Despite popular narratives about an era of data abundance, there is a chronic lack of scientific data on most of the natural and social world, from the soil, deep ocean and atmosphere, to the informal economy. AI could help in different ways. It could make existing data collection more accurate, for example by reducing the noise and errors

that can occur when [sequencing DNA](#), [detecting cell types in a sample](#), or [capturing animal sounds](#). Scientists can also exploit LLMs' growing ability to operate across images, video and audio, to extract the *unstructured* scientific data that is buried in scientific publications, archives, and less obvious resources such as instructional videos, and convert it into structured datasets.

尽管关于数据丰富的时代有很多流行叙事，但大多数自然和社会领域，从土壤、深海和大气，到非正规经济，都存在科学数据的长期匮乏。人工智能可以从不同方面提供帮助。例如，它可以提高现有数据收集的准确性，通过减少在测序 DNA、检测样本中的细胞类型或捕捉动物声音时可能出现的噪声和错误。科学家还可以利用 LLMs 日益增长的能力，跨图像、视频和音频进行操作，从科学出版物、档案以及不太明显的资源（如教学视频）中提取非结构化科学数据，并将其转换为结构化数据集。

AI can also help to annotate scientific data with the supporting information that scientists need in order to use it. For example, at least one-third of microbial proteins [are not reliably annotated](#) with details about the function(s) that they are thought to perform. In 2022, our researchers [used AI to predict the function of proteins](#), leading to new entries in the [UniProt](#), [Pfam](#) and [InterPro](#) databases. AI models, once validated, can also serve as new sources of synthetic scientific data. For example, our [AlphaProteo](#) protein design model is trained on more than 100 million AI-generated protein structures from AlphaFold 2, along with experimental structures from the [Protein Data Bank](#). These AI opportunities can complement and increase the return on other much-needed efforts to generate scientific data, such as [digitising archives](#), or funding new data capture technologies and methods, like efforts underway in single cell genomics to create powerful datasets of individual cells in unprecedented detail.

AI 也能帮助科学家标注科学数据，并提供他们使用这些数据所需的支持信息。例如，至少有三分之一微生物蛋白的功能细节标注不可靠。2022 年，我们的研究人员利用 AI 预测蛋白质功能，从而在 UniProt、Pfam 和 InterPro 数据库中新增了条目。经过验证的 AI 模型还可以作为新的合成科学数据来源。例如，我们的 AlphaProteo 蛋白质设计模型在训练时使用了超过 1 亿个由 AlphaFold 2 生成的 AI 蛋白质结构，以及来自蛋白质数据库的实验结构。这些 AI 机遇可以补充并提高其他生成科学数据的必要工作的回报，例如数字化档案，或资助新的数据捕获技术和方法，如单细胞基因组学中正在进行的努力，以创建前所未有的详细程度的单个细胞强大数据集。

3. EXPERIMENTS: Simulate, accelerate and inform complex experiments

3. 实验：模拟、加速和指导复杂实验

Many scientific experiments are expensive, complex, and slow. Some do not happen at all because researchers cannot access the facilities, participants or inputs that they need. Fusion is a case in point. It promises an energy source that is practically limitless, emission-free and could enable the scaling of energy-intensive innovations, like desalination. To realise fusion, scientists need to create and control *plasma* - a fourth fundamental state of matter. However, the facilities needed are hugely complex to build. ITER's prototype tokamak reactor began construction in 2013, but plasma experiments are [not set to begin](#) until the mid-2030s at the earliest, although others hope to build smaller reactors on shorter timelines.

许多科学实验既昂贵又复杂，而且耗时很长。有些实验甚至根本无法进行，因为研究人员无法获得所需的设备、参与者或输入。聚变就是一个典型的例子。它承诺提供一种几乎无限的、无排放的能源，并能够推动高能耗创新技术的规模化发展，例如海水淡化。要实现聚变，科学家需要创造并控制等离子体——物质的一种第四种基本状态。然而，所需的设备极其复杂，建造难度很大。国际热核聚变实验堆（ITER）的原型托卡马克反应堆于 2013 年开始建设，但等离子体实验最早也要等到 2030 年代中期才能开始，尽管其他人希望能在更短的时间表内建造小型反应堆。

AI could help to simulate fusion experiments and enable much more efficient use of subsequent experiment time. One approach is to run reinforcement learning agents on simulations of physical systems. Between 2019 and 2021, our researchers partnered with the Swiss Federal Institute of Technology Lausanne to [demonstrate](#) how to use RL to control the shape of plasma in a simulation of a tokamak reactor. These approaches could be extended to other experimental facilities, such as [particle accelerators](#), [telescope arrays](#), or [gravitational wave detectors](#).

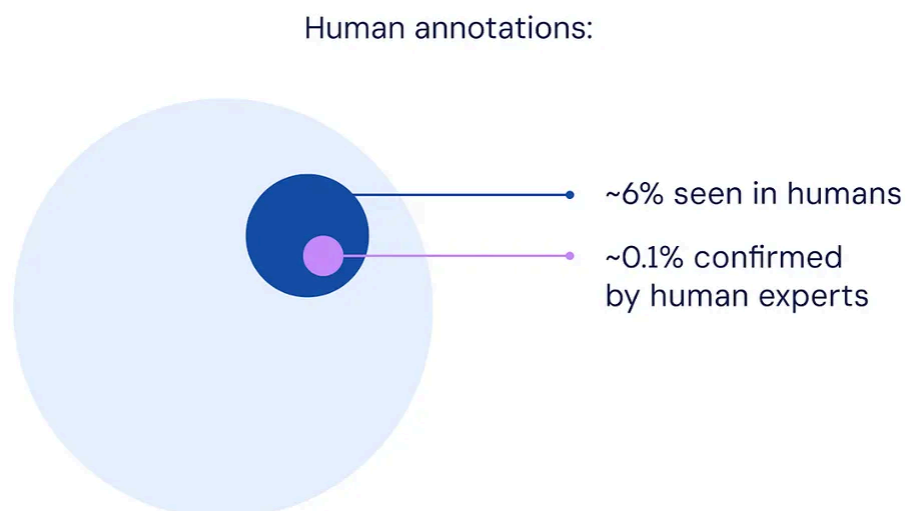
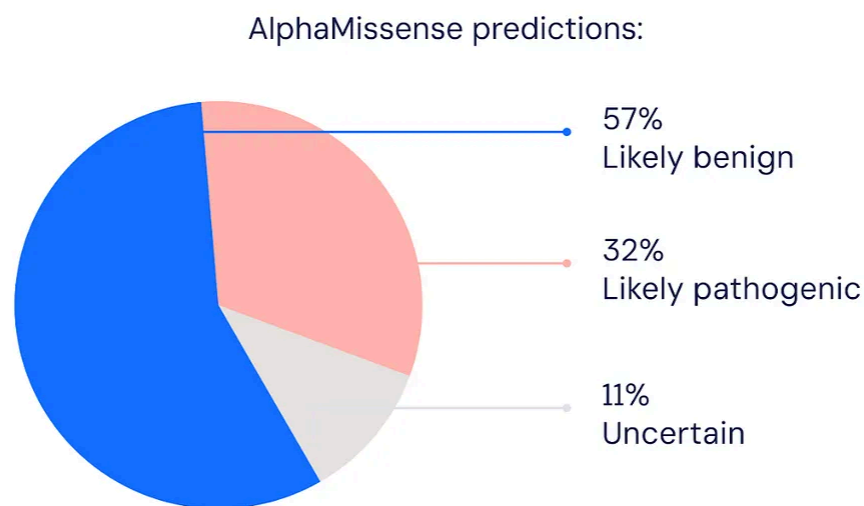
AI 可以帮助模拟核聚变实验，并使后续实验时间得到更高效的利用。一种方法是运行强化学习智能体在物理系统的模拟上。在 2019 年至 2021 年期间，我们的研究人员与瑞士洛桑联邦理工学院合作，展示了如何使用强化学习来控制托卡马克反应堆模拟中等离子体的形状。这些方法可以扩展到其他实验设施，例如粒子加速器、望远镜阵列或引力波探测器。

Using AI to simulate experiments will look very different across disciplines, but a common thread is that the simulations will often inform and guide physical experiments, rather than substitute for them. For example, the average person [has more than 9,000 missense variants](#), or single letter substitutions in their DNA. Most of these genetic variants are benign but some can disrupt the functions that proteins perform, contributing to rare genetic diseases like cystic fibrosis as well as common

diseases like cancer. Physical experiments to test the effects of these variants are often limited to a single protein. Our [AlphaMissense model](#) classifies 89% of the 71 million potential human missense variants as likely harmful or benign, enabling scientists to focus their physical experiments on the most likely contributors to disease.

使用 AI 模拟实验在不同学科中会呈现出很大的差异，但共同点是模拟实验往往能指导和补充物理实验，而不是完全替代它们。例如，普通人的 DNA 中有多于 9,000 种错义变异，即单个字母的替换。这些遗传变异大多数是无害的，但有些会破坏蛋白质的功能，导致囊性纤维化等罕见遗传病以及癌症等常见病。测试这些变异对蛋白质影响的物理实验通常仅限于单个蛋白质。我们的 AlphaMissense 模型将 7100 万种潜在人类错义变异中的 89% 分类为可能有害或无害，使科学家能够将物理实验集中在最可能导致疾病的变异上。

All possible 71 million human missense variants



AlphaMissense predicted the pathogenicity of all possible 71 million missense variants. It classified 89% - predicting 57% were likely benign and 32% were

4. MODELS: Model complex systems and how their components interact

4. 模型：模拟复杂系统及其组成部分的相互作用

In a [1960 paper](#), the Nobel Prize winning physicist Eugene Wigner marvelled at the "unreasonable effectiveness" of mathematical equations for modelling important natural phenomena, such as planetary motion. However, over the past half century, models that rely on sets of equations or other deterministic assumptions have struggled to capture [the full complexity of systems in biology, economics, weather, and elsewhere](#). This reflects the sheer number of interacting parts that make up these systems, as well as their dynamism and potential for emergent, random or chaotic behaviour. The challenges in modelling these systems impedes scientists' ability to predict or control how they will behave, including during shocks or interventions, such as rising temperatures, a new drug, or the introduction of a tax change.

在 1960 年的一篇论文中，诺贝尔奖获得者、物理学家尤金·维格纳惊叹于数学方程在模拟重要自然现象（如行星运动）时的“不可思议的有效性”。然而，在过去的半个世纪里，依赖方程组或其他确定性假设的模型难以捕捉生物学、经济学、天气等领域的系统全部复杂性。这反映了这些系统由大量相互作用的部分构成，以及它们的动态性和涌现、随机或混沌行为的可能性。对这些系统建模的挑战阻碍了科学家预测或控制它们行为的能力，包括在温度上升、新药物推出或税收政策变化等冲击或干预期间。

[AI could more accurately model these complex systems](#) by ingesting more data about them, and learning more powerful patterns and regularities within this data. For example, modern weather forecasting is a triumph of science and engineering. For governments and industry, it informs everything from renewable energy planning to preparing for [hurricanes](#) and [floods](#). For the public, the weather is the most popular non-branded query on Google Search. Traditional [numeral prediction methods](#) are based on carefully-defined physics equations that provide a very useful, yet imperfect, approximation of the atmosphere's complex dynamics. They are also computationally expensive to run. In 2023, we released a [deep learning system](#) that predicts weather conditions up to 10 days in advance, which outperformed traditional models on accuracy and prediction speed. As we expand on below, using AI to forecast weather variables could also help to mitigate and respond to climate change. For instance, when pilots fly through humid regions it can cause *condensation trails* that [contribute](#)

to aviation's global warming impact. Google scientists [recently used](#) AI to predict when and where humid regions may arise to help pilots avoid flying through them.

AI 可以通过摄入更多关于这些复杂系统的数据，从而更准确地模拟它们，并在这些数据中学习更强大的模式和规律。例如，现代天气预报是科学与工程巨大胜利。对于政府和工业来说，它影响着从可再生能源规划到准备应对飓风和洪水等方方面面。对于公众而言，天气是谷歌搜索中最受欢迎的非品牌查询。传统的数值预测方法基于精心定义的物理方程，这些方程为大气复杂的动力学提供了非常有用但并不完美的近似。它们运行起来也需要很高的计算成本。2023 年，我们发布了一个深度学习系统，可以提前 10 天预测天气状况，在准确性和预测速度上均优于传统模型。正如我们下面将详细阐述的，使用 AI 预测天气变量还有助于缓解和应对气候变化。例如，当飞行员飞越潮湿地区时，可能会形成凝结尾迹，这会加剧航空对全球变暖的影响。谷歌科学家最近使用人工智能预测了潮湿地区可能出现的时机和地点，以帮助飞行员避免飞越这些区域。

In many cases, AI will enrich traditional approaches to modelling complex systems rather than replace them. For example, agent-based modelling simulates interactions between individual actors, like firms and consumers, to understand how these interactions might affect a larger more complex system like the economy. Traditional approaches require scientists to specify beforehand how these computational agents should behave. Our research teams [recently outlined](#) how scientists could use LLMs to create more flexible *generative* agents that communicate and take actions, such as searching for information or making purchases, while also reasoning about and remembering these actions. Scientists could also use reinforcement learning to study how these agents learn and adapt their behaviour in [more dynamic simulations](#), for example in response to the introduction of new energy prices or pandemic response policies.

在许多情况下，人工智能将丰富对复杂系统建模的传统方法，而不是取代它们。例如，基于主体的建模通过模拟个体行为者（如企业和消费者）之间的互动，来理解这些互动如何影响更大的复杂系统（如经济）。传统方法要求科学家事先指定这些计算主体的行为方式。我们近期的研究团队概述了科学家如何使用 LLMs 创建更灵活的生成式主体，这些主体能够进行交流并采取行动，例如搜索信息或进行购买，同时还能推理和记忆这些行动。科学家还可以使用强化学习来研究这些主体如何在更动态的模拟中学习和调整其行为，例如在新能源价格引入或疫情应对政策实施时的反应。

5. SOLUTIONS: Identify novel solutions to problems with large search spaces

5. 解决方案：为具有大搜索空间的问题寻找新颖的解决方案

Many important scientific problems come with a practically incomprehensible number of potential solutions. For example, biologists and chemists aim to determine the structure, characteristics, and function(s) of molecules such as proteins. One goal of such work is to help design novel versions of these molecules to serve as antibody drugs, plastic-degrading enzymes or new materials. However, to design a small molecule drug, scientists face [more than \$10^{60}\$](#) potential options. To design a protein with 400 standard amino acids, they face 20^{400} options. These large search spaces are not limited to molecules but are commonplace for many scientific problems, such as finding the best proof for a maths problem, the most efficient algorithm for a computer science task, or the [best architecture for a computer chip](#).

许多重要的科学问题都伴随着实际上难以理解的潜在解决方案数量。例如，生物学家和化学家旨在确定蛋白质等分子的结构、特性以及功能。这类工作的一个目标是帮助设计这些分子的新版本，以用作抗体药物、塑料降解酶或新材料。然而，要设计一种小分子药物，科学家面临超过 10^{60} 种潜在选择。要设计一种含有 400 种标准氨基酸的蛋白质，他们面临 20^{400} 种选择。这些大搜索空间不仅限于分子，在许多科学问题中也普遍存在，例如为数学问题寻找最佳证明、为计算机科学任务寻找最高效的算法，或为计算机芯片寻找最佳架构。

Traditionally, scientists rely on some combination of intuition, trial and error, iteration, or brute force computing to find the best molecule, proof, or algorithm. However, these methods struggle to exploit the huge space of potential solutions, leaving better ones undiscovered. AI can [open up new parts of these search spaces](#) while also homing in more quickly on the solutions that are most likely to be viable and useful - a delicate balancing act. For example, in July, our [AlphaProof and AlphaGeometry 2](#) systems correctly solved four out of six problems from the [International Mathematical Olympiad](#), an elite high school competition. The systems make use of our Gemini LLM architecture to generate a large number of novel ideas and potential solutions for a given maths problem, and combine this with systems grounded in mathematical logic that can iteratively work towards the candidate solutions that are most likely to be

correct.

传统上，科学家依赖直觉、试错、迭代或蛮力计算来寻找最佳分子、证明或算法。然而，这些方法难以利用庞大的潜在解空间，导致更好的解被遗漏。人工智能可以打开这些搜索空间的新部分，同时更快地锁定最有可能可行和有用的解——这是一项精妙的平衡工作。例如，今年 7 月，我们的 AlphaProof 和 AlphaGeometry 2 系统正确解决了国际数学奥林匹克竞赛（一项精英高中竞赛）中的六个问题中的四个。这些系统利用我们的 Gemini LLM 架构为给定的数学问题生成大量新颖想法和潜在解，并结合基于数学逻辑的系统，这些系统可以迭代地工作以接近最有可能正确的候选解。

AI scientists or AI-empowered scientists?

AI 科学家还是 AI 赋能的科学家？

This growing use of AI in science, and the emergence of early AI scientific assistants, raises questions about how fast and how far the capabilities of AI may advance and what this will mean for human scientists. Current LLM-based AI scientific assistants make a relatively small contribution to a relatively narrow range of tasks, such as supporting literature reviews. There are plausible near-term scenarios in which they become better at these tasks and become capable of more impactful ones, such as helping to generate powerful hypotheses, or helping to predict the outcomes of experiments. However, current systems still struggle with the deeper creativity and [reasoning](#) that human scientists rely on for such tasks. [Efforts are underway to improve these AI capabilities](#), for example by combining LLMs with logical deduction engines, as in our [AlphaProof and AlphaGeometry 2](#) examples, but further breakthroughs are needed. The ability to accelerate or automate experiments will also be harder for those that require complicated actions in wet labs, interacting with human participants, or lengthy processes, such as monitoring disease progression. Although again, work is underway in some of these areas, such as new types of

laboratory robotics and [automated labs](#).

AI 在科学领域的日益广泛使用，以及早期 AI 科学助手的涌现，引发了关于 AI 能力可能发展速度和范围的问题，以及这对人类科学家意味着什么。当前的基于 LLM 的 AI 科学助手对相对狭窄的任务范围（如支持文献综述）做出相对较小的贡献。在近期可能出现的合理场景中，它们在这些任务上表现会更好，并能够承担更具影响力的任务，例如帮助生成强大的假设，或帮助预测实验结果。然而，当前系统在人类科学家依赖的更深层次的创造力和推理方面仍存在困难。人们正在努力提升这些 AI 能力，例如通过将 LLM 与逻辑推理引擎结合，正如我们的 AlphaProof 和 AlphaGeometry 2 示例所示，但仍需进一步突破。对于需要在湿实验室进行复杂操作、与人类参与者互动或经历漫长过程的实验（如监测疾病进展），加速或自动化实验的能力也将更加困难。尽管如此，在这些领域中的某些地方，例如新型实验室机器人和自动化实验室，工作正在进行中。

Even as AI systems' capabilities improve, the greatest marginal benefit will come from deploying them in use cases that play to their relative strengths - such as the ability to rapidly extract information from huge datasets - and which help address genuine bottlenecks to scientific progress such as the five opportunities outlined above, rather than automating tasks that human scientists already do well. As AI enables cheaper and more powerful science, demand for science and scientists will also grow. For example, recent breakthroughs have already led to a slew of new startups in areas like [protein design](#), [material science](#) and [weather forecasting](#). Unlike other sectors, and despite [past claims to the contrary](#), future demand for science appears practically limitless. New advances [have always opened up new, unpredictable regions](#) in the scientific *map of knowledge*, and AI will do similar. As [envisioned](#) by Herbert Simon, AI systems will also become *objects* of science research, with scientists set to play a leading role in evaluating and explaining their scientific capabilities, as well as in developing new types of human-AI scientific systems.

随着 AI 系统能力的提升，最大的边际效益将来自于将其部署在使用案例中，这些案例能够发挥其相对优势——例如快速从海量数据集中提取信息——并帮助解决科学进步的真实瓶颈，如上述五个机遇，而不是自动化人类科学家已经做得很好的任务。随着 AI 使科学变得更便宜、更强大，对科学和科学家的需求也将增长。例如，最近的突破已经导致了在蛋白质设计、材料科学和天气预报等领域出现了一系列新创业公司。与其他领域不同，尽管过去有人提出相反观点，但未来对科学的需求实际上几乎是无限的。新的进展总是会在知识科学地图上开辟出新的、不可预测的区域，AI 也将类似地做到这一点。正如赫伯特·西蒙所设想的那样，AI 系统也将成为科学研究的研究对象，科学家将在评估和解释其科学能力方面发挥主导作用，同时在新类型的人机科学系统中发挥作用。

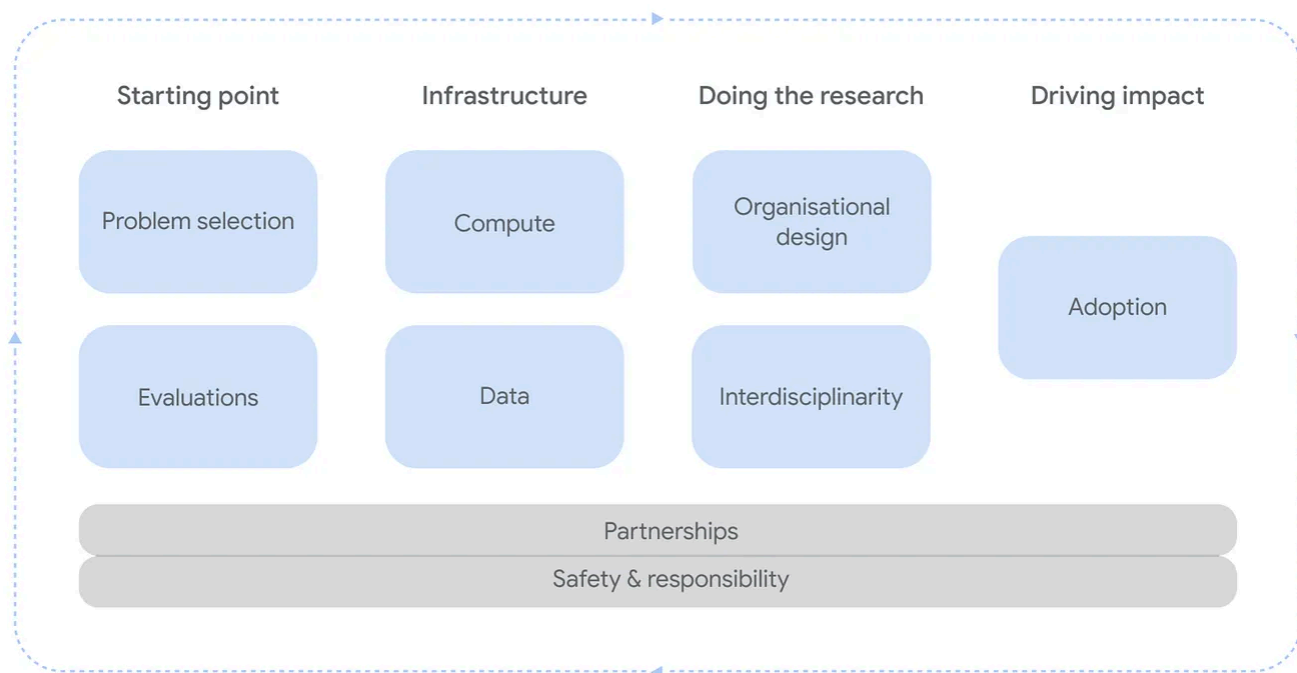
Part B. The ingredients

B 部分。成分

We are interested in the ingredients that ambitious AI for Science efforts need to succeed - both at the individual research effort level and at the level of the science ecosystem, where policymakers have more scope to shape them. The experts that we interviewed routinely cited several ingredients that we organised into a toy model, which we call *the AI for Science production function*. This production function is not meant to be exhaustive, prescriptive, or a neat linear process. The ingredients will be intuitive to many, but our interviews revealed a number of lessons about what they look like in practice which we share below.

我们对雄心勃勃的 AI 科学项目成功所需的要素感兴趣——这既包括个人研究层面的要素，也包括科学生态层面的要素，后者为政策制定者提供了更大的塑造空间。我们采访的专家们经常提到几个要素，我们将这些要素组织成一个简单的模型，称之为 AI 科学生产函数。这个生产函数并非旨在详尽无遗、具有指导性或是一个简洁的线性过程。这些要素对许多人来说可能很直观，但我们的采访揭示了许多关于它们在实践中具体表现的经验教训，我们在此分享。

The AI for science production function



1. PROBLEM SELECTION: Pursue ambitious, AI-shaped problems

1. 问题选择：追求雄心勃勃的、由 AI 塑造的问题

Scientific progress rests on being able to identify an important problem and ask the *right* question about how to solve it. In [their exploration](#) into the genesis of scientific breakthroughs, Venkatesh Narayanamurti and Jeffrey Y. Tsao document how important the reciprocal and recursive relationship between questions and answers is, including the importance of asking ambitious new questions. Our Science team starts by thinking about whether a potential research problem is significant enough to justify a substantial investment of time and resources. Our CEO Demis Hassabis has a mental model to guide this assessment: thinking about all of science as a *tree of knowledge*. We are particularly interested in the roots - fundamental 'root node problems' like [protein structure prediction](#) or [quantum chemistry](#) that, if solved, could unlock entirely new branches of research and applications.

科学进步依赖于能够识别重要问题并就如何解决它提出正确的问题。在探索科学突破的起源时，Venkatesh Narayanamurti 和 Jeffrey Y. Tsao 记录了问题与答案之间互惠和递归关系的重要性，包括提出雄心勃勃的新问题的重要性。我们的科学团队首先思考潜在的研究问题是否足够重要，值得投入大量时间和资源。我们的 CEO Demis Hassabis 有一个思维模型来指导这一评估：将所有科学视为知识之树。我们对树根特别感兴趣——基础“根节点问题”，如蛋白质结构预测或量子化学，如果解决这些问题，将能解锁全新的研究分支和应用。

To assess whether AI will be suitable and additive, we look for problems with certain characteristics, such as huge combinatorial search spaces, large amounts of data, and a clear objective function to benchmark performance against. Often a problem is suitable for AI in principle, but the inputs aren't yet in place and it needs to be stored for later. One of the original [inspirations](#) for AlphaFold was conversations that Demis had many years prior as a student with a friend who was obsessed with the protein folding problem. Many recent breakthroughs also feature this coming together of an important scientific problem and an AI approach that has reached a point of maturity. For example, our [fusion](#) effort was aided by a novel [reinforcement learning algorithm](#) called *maximum a posteriori policy optimization*, which had only just been released. Alongside a new fast and [accurate simulator](#) that our partners EPFL had just

developed, that enabled the team to overcome a data paucity challenge.

要评估 AI 是否适用且能带来增益，我们会寻找具有特定特征的问题，例如巨大的组合搜索空间、大量数据以及明确的性能基准目标函数。通常，一个问题在原则上适合 AI，但输入数据尚未就绪，需要存储以备后用。AlphaFold 最初的灵感之一，源于多年前 Demis 作为一名学生与一位痴迷于蛋白质折叠问题的朋友之间的对话。许多最近的突破也体现了重要科学问题与成熟 AI 方法的结合。例如，我们的融合工作得到了一种名为最大后验策略优化的新型强化学习算法的帮助，该算法刚刚发布。再加上我们合作伙伴 EPFL 刚刚开发的新型快速准确模拟器，使团队能够克服数据匮乏的挑战。

In addition to picking the right problem, it is important to *specify* it at the right level of difficulty. Our interviewees emphasised that a powerful problem statement for AI is often one that lends itself to *intermediate* results. If you pick a problem that's too hard then you won't generate enough signal to make progress. Getting this right relies on intuition and experimentation.

除了选择正确的问题，在适当的难度级别上明确问题同样重要。我们的受访者强调，对于人工智能来说，一个强有力的问题描述往往适合产生中间结果。如果你选择一个太难的问题，那么你将无法产生足够的信号来取得进展。要正确把握这一点，需要依靠直觉和实验。

2. EVALUATIONS: Invest in evaluation methods that can provide a robust performance signal and are endorsed by the community

2. 评估：投资于能够提供稳健性能信号且得到社区认可的评价方法

Scientists use evaluation methods, such as benchmarks, metrics and competitions, to assess the scientific capabilities of an AI model. Done well, these evaluations provide a way to track progress, encourage innovation in methods, and galvanise researchers' interest in a scientific problem. Often, a variety of evaluation methods are required. For example, our [weather forecasting](#) team started with an initial 'progress metric' based on a few key variables, such as surface temperature, that they used to 'hill climb', or gradually improve their model's performance. When the model had reached a certain level of performance, they carried out a more comprehensive evaluation using more than 1,300 metrics inspired by the [European Centre for Medium-Range Weather Forecasts's](#) evaluation [scorecard](#). In past work, the team learned that AI models can sometimes achieve good scores on these metrics in undesirable ways. For example, 'blurry' predictions - such as predicting rainfall within a large geographical area - are less penalised than 'sharp' predictions - such as predicting a storm in a location that is

very slightly different to the actual location - the so-called 'double-penalty' problem. To provide further verification, the team evaluated the usefulness of their model on downstream tasks, such as its ability to predict the track of a cyclone, and to characterise the strength of 'atmospheric rivers' - narrow bands of concentrated moisture that can lead to flooding.

科学家使用基准测试、指标和竞赛等评估方法来衡量人工智能模型的科学能力。如果执行得当，这些评估提供了一种跟踪进展、鼓励方法创新并激发研究人员对科学问题兴趣的方式。通常，需要多种评估方法。例如，我们的天气预报团队最初基于几个关键变量（如地表温度）设定了一个初始的“进度指标”，用于“爬山”，即逐步提高模型性能。当模型达到一定性能水平后，他们使用欧洲中期天气预报中心评估记分卡启发的 1300 多个指标进行了更全面的评估。在以往的工作中，团队发现人工智能模型有时会以不理想的方式在这些指标上获得高分。例如，“模糊”预测（如在广大地理区域内预测降雨）受到的惩罚比“锐利”预测（如在非常轻微偏离实际位置的地方预测风暴）要轻——这就是所谓的“双重惩罚”问题。为进行进一步验证，团队评估了他们的模型在下游任务中的实用性，例如预测飓风路径的能力，以及表征“大气河流”——这些集中了水分的狭窄带状区域可能导致洪水——的强度。

The most impactful AI for Science evaluation methods are often community-driven or endorsed. A gold standard is the [Critical Assessment of protein Structure Prediction](#) competition. Established in 1994 by [Professor John Moult and Professor Krzysztof Fidelis](#), the biennial CASP competition has challenged research groups to test the accuracy of their protein structure prediction models against real, unreleased, experimental protein structures. It has also become a unique global community and a catalyst for research progress, albeit one that is hard to replicate quickly. The need for community buy-in also provides an argument for publishing benchmarks so that researchers can use, critique and improve them. However, this also creates the risk that the benchmark will 'leak' into an AI model's training data, reducing its usefulness for tracking progress. There is no perfect solution to this tradeoff but, at a minimum, new public benchmarks are needed at regular intervals. Scientists, AI labs and policymakers should also explore new ways to assess the scientific capabilities of AI models, such as setting up new third-party assessor organisations, competitions, and

enabling more open-ended probing of AI models' capabilities by scientists.

对科学最有影响力的 AI 评估方法通常是社区驱动或认可的。一个黄金标准是蛋白质结构预测的批判性评估竞赛。由约翰·穆尔特教授和克日什托夫·菲德利斯教授于 1994 年创立的这项每两年举办一次的 CASP 竞赛，挑战研究团队用真实且未发布的实验蛋白质结构来测试其蛋白质结构预测模型的准确性。它已成为一个独特的全球社区和研究进展的催化剂，尽管很难快速复制。社区认可的需求也为发布基准测试提供了理由，以便研究人员可以使用、批评和改进它们。然而，这也带来了基准测试可能“泄露”到 AI 模型的训练数据中的风险，从而降低其跟踪进度的有效性。在这个权衡中没有完美的解决方案，但至少需要定期发布新的公共基准测试。科学家、人工智能实验室和政策制定者还应该探索评估人工智能模型科学能力的新方法，例如建立新的第三方评估机构、举办竞赛，并让科学家能够更开放地探究人工智能模型的能力。

3. COMPUTE: Track how compute use is evolving and invest in specialist skills

3. 计算：追踪计算使用的演变情况，并投资于专业技能

Multiple [government reviews have recognised](#) the growing importance of compute to progress in AI, science, and the wider economy. As we expand on further below, there is also a growing focus on its energy consumption and greenhouse gas emissions. AI labs and policymakers should take a grounded, long-term view that considers how compute needs will vary across AI models and use cases, potential multiplier effects and efficiency gains, and how this compares to counterfactual approaches to scientific progress that don't use AI.

多个政府审查已认识到计算能力对人工智能、科学和更广泛经济进步日益增长的重要性。正如我们下面将进一步阐述的，人们也越来越关注其能源消耗和温室气体排放。人工智能实验室和政策制定者应采取基于现实的、长远的视角，考虑计算需求将如何在不同的人工智能模型和应用场景中变化，潜在的乘数效应和效率提升，以及这与不使用人工智能的科学进步的对比。

For example, some state-of-the-art AI models, such as in protein design, are relatively small. Larger models, like LLMs, are compute-intensive to train but typically require much less compute to fine-tune, or to run inference against, which can open up more efficient pathways to science research. Once an LLM is trained, it is also easier to make it more efficient, for example via better [data curation](#), or by '[distilling](#)' the large model into a smaller one. AI compute needs should also be evaluated in comparison to other models of scientific progress. For example, AI weather forecasting models are compute-intensive to train, but [can still be](#) more computationally-efficient than

traditional techniques. These nuances highlight the need for AI labs and policymakers to track compute use empirically, to understand how it is evolving, and to project what these trends mean for future demand. In addition to ensuring sufficient access to the right kind of chips, a compute strategy should also prioritise the critical infrastructure and engineering skills needed to manage access and ensure reliability. This is often under-resourced in academia and public research institutions.

例如，一些处于顶尖水平的 AI 模型，如蛋白质设计领域的模型，相对较小。像 LLMs 这样的大型模型训练计算密集，但通常微调或进行推理所需的计算量要少得多，这可以开辟更高效的科研途径。一旦训练好 LLM，也更容易使其更高效，例如通过更好的数据管理，或通过将大型模型“蒸馏”成小型模型。AI 计算需求也应该与其他科学进步模型进行比较评估。例如，AI 天气预测模型训练计算密集，但仍然可能比传统技术更计算高效。这些细微之处突显了 AI 实验室和政策制定者需要通过实证追踪计算使用情况、了解其演变方式，以及预测这些趋势对未来需求意味着什么的必要性。除了确保获得合适的芯片，计算策略还应优先考虑管理访问和确保可靠性的关键基础设施和工程技能。这在学术界和公共研究机构中往往资源不足。

4. DATA: Blend top-down and bottom-up efforts to collect, curate, store, and access data

4. 数据：结合自上而下和自下而上的方法，收集、整理、存储和访问数据

Similar to compute, data can be viewed as critical *infrastructure* for AI for Science efforts that needs to be developed, maintained, and updated over time. Discussions often focus on identifying new datasets that policymakers and practitioners should create. There is a role for such top-down efforts. In 2012, the Obama Administration launched the [Materials Project](#) to map known and predicted materials, such as inorganic crystals, like silicon, that are found in batteries, solar panels, and computer chips. Our recent [GNoME](#) effort used this data to predict 2.2 million novel inorganic crystals, including 380,000 that simulations suggest are stable at low temperatures, making them candidates for new materials.

与计算类似，数据可以被视为需要随着时间推移而开发、维护和更新的科学 AI 关键基础设施。讨论往往集中在识别政策制定者和从业者应创建的新数据集上。这种自上而下的努力有其作用。2012 年，奥巴马政府启动了材料项目，旨在绘制已知和预测的材料，例如无机晶体（如硅），这些材料存在于电池、太阳能电池板和计算机芯片中。我们最近的 GNoME 项目利用这些数据预测了 220 万种新型无机晶体，其中包括 38 万种模拟表明在低温下稳定的晶体，使其成为新材料的候选者。

However, it is often difficult to predict in advance what scientific datasets will be most important, and many AI for Science breakthroughs rely on data that emerged more organically, thanks to the efforts of an enterprising individual or small teams. For example, [Daniel MacArthur](#), then a researcher at the Broad Institute, led the development of the [gnomAD](#) dataset of genetic variants that our [AlphaMissense](#) work subsequently drew on. Similarly, the mathematical proof assistant and programming language [Lean](#) was originally developed by the programmer [Leonardo de Moura](#). It is not a dataset, but many AI labs now use it to help train their AI maths models, including our [AlphaProof](#) system.

然而，预先预测哪些科学数据集将最为重要往往很困难，许多 AI for Science 的突破依赖于那些因个人创业者或小团队的努力而更自然产生的数据。例如，丹尼尔·马卡瑟（当时是 Broad 研究所的研究员），领导开发了 gnomAD 基因变异数据集，我们的 AlphaMissense 工作随后便借鉴了该数据集。类似地，数学证明助手和编程语言 Lean 最初由程序员莱昂纳多·德·莫拉开发。它不是数据集，但许多 AI 实验室现在使用它来帮助训练他们的 AI 数学模型，包括我们的 AlphaProof 系统。

Efforts like [gnomAD](#) or [Lean](#) highlight how top-down data efforts need to be complemented by better incentives for individuals at all stages of the data pipeline. For example, some data from strategic wet lab experiments is currently discarded, but could be collected and stored, if stable funding was available. Data curation could also be better incentivised. Our [AlphaFold](#) models were trained on data from the [Protein Data Bank](#) that was particularly high quality because journals require the deposition of protein structures as a precondition for publication, and the PDB's professional data curators developed standards for this data. In genomics, many researchers are also obliged to deposit raw sequencing data in the [Sequence Read Archive](#) but inconsistent standards mean that individual datasets often still need to be reprocessed and combined. Some other high-quality datasets go unused altogether, because of restrictive licensing conditions, such as in biodiversity, or because the datasets are not released, such as decades of data from publicly-funded fusion experiments. There can be logical reasons for this, such as a lack of time, funds, somewhere to put the data, or the need for temporary embargo periods for researchers who develop the data. But in aggregate these data access issues pose a

key bottleneck to using AI to advance scientific progress.

像 gnomAD 或 Lean 这样的工作突显了自上而下的数据工作需要通过更好的激励机制来补充数据管道中各个阶段个人的参与。例如，目前一些来自战略性湿实验的数据被丢弃了，但如果能提供稳定资金，这些数据本可以收集和存储。数据管理也可以通过更好的激励机制来促进。我们的 AlphaFold 模型是在来自蛋白质数据库的数据上训练的，这些数据质量特别高，因为期刊要求将蛋白质结构作为发表的前提条件进行提交，而蛋白质数据库的专业数据管理者为这些数据制定了标准。在基因组学领域，许多研究人员也必须将原始测序数据提交到序列读取存档库，但由于标准不一致，单个数据集往往仍需重新处理和整合。一些其他高质量的数据集完全未被使用，原因在于限制性的许可条件，例如在生物多样性领域，或者因为数据集未被发布，例如来自公共资助的聚变实验数十年的数据。这种情况下可能有合理的逻辑原因，例如缺乏时间、资金、数据存放之处，或需要为开发数据的研究人员设置临时禁令期。但总体而言，这些数据访问问题构成了使用人工智能推动科学进步的关键瓶颈。

5. ORGANISATIONAL DESIGN: Strike the right balance between bottom-up creativity and top-down coordination

5. 组织设计：在自下而上的创造力和自上而下的协调之间取得平衡

A simple heuristic is that academia and industry tend to approach science research at two ends of a spectrum. Academia tends to be more bottom-up, and industry labs tend to be more top-down. In reality, there has long been plenty of space in between, particularly at the most successful labs, such as the golden eras of Bell Labs and Xerox PARC that were renowned for their blue skies research and served as inspiration in the founding of DeepMind. Recently, a new wave of [science research institutions](#) has emerged that try to [learn from](#) these outlier examples. These organisations differ in their goals, funding models, disciplinary focus, and how they organise their work. But collectively they want to deliver more high-risk, high-reward research, less bureaucracy, and better incentives for scientists. Many have a strong focus on applying AI, such as the UK's [Advanced Research and Invention Agency](#), the [Arc Institute](#), and the growing number of [Focused Research Organisations](#) that aim to tackle specific problems in science that are too large for academia and not profitable enough for industry, such as the [organisation](#) tasked with expanding the Lean proof

assistant that is pivotal to AI maths research.

一个简单的经验法则是，学术界和工业界往往以光谱的两端来处理科学研究。学术界倾向于更自下而上，而工业实验室则倾向于更自上而下。实际上，两者之间一直存在很大的空间，尤其是在最成功的实验室中，例如以蓝天空研究而闻名的贝尔实验室和施乐 PARC 黄金时代，它们为 DeepMind 的成立提供了灵感。最近，出现了一批新的科学研究机构，它们试图从这些极端案例中学习。这些组织在目标、资金模式、学科重点和工作组织方式上有所不同。但总体而言，它们都希望开展更多高风险、高回报的研究，减少官僚主义，并为科学家提供更好的激励。许多人专注于应用 AI，例如英国的先进研究与发明署、Arc 研究所，以及越来越多的专注研究组织，这些组织旨在解决科学中那些规模太大不适合学术界、而利润又不足以吸引工业界的问题，比如负责扩展对 AI 数学研究至关重要的 Lean 证明助手组织的任务。

At their core, these new institutions share a desire to find a better blend of top-down coordination and focus with bottom-up empowerment of scientists. For some organisations, this means focussing on a single specific problem with pre-specified milestones. For others, it means offering more unrestricted funding to principal investigators. Getting this balance right is critical to attracting and retaining research leaders, who must also buy into it if it is to succeed - Demis Hassabis has credited it as the single biggest factor for successfully coordinating cutting-edge research at scale. Striking this balance is also important *within* individual research efforts. In Google DeepMind's case, efforts often pivot between more unstructured 'exploration' phases, where teams search for new ideas, and faster 'exploitation' phases, where they focus on engineering and scaling performance. There is an art to knowing when to switch between these modes and how to adapt the project team accordingly.

这些新机构的本质都渴望找到自上而下的协调与专注和自下而上的科学家赋权之间的更好平衡。对某些组织而言，这意味着专注于一个特定的具体问题，并设定预定的里程碑。对另一些组织而言，这意味着为首席研究员提供更多不受限制的资金。正确地把握这种平衡对于吸引和留住研究领导者至关重要，他们必须认同这一点，否则就无法成功——Demis Hassabis 将其视为成功大规模协调尖端研究的最大因素。在个人研究工作中，找到这种平衡也同样重要。在 Google DeepMind 的案例中，工作往往在更无结构的“探索”阶段之间切换，在这些阶段，团队寻找新想法，以及更快的“利用”阶段，在这些阶段，他们专注于工程和扩展性能。知道何时在这些模式之间切换以及如何相应地调整项目团队是一门艺术。

6. INTERDISCIPLINARITY: Approach science as a team, fund neglected roles, and promote a culture of contestability

6. 跨学科合作：以团队方式对待科学，资助被忽视的领域，并促进竞争性文化

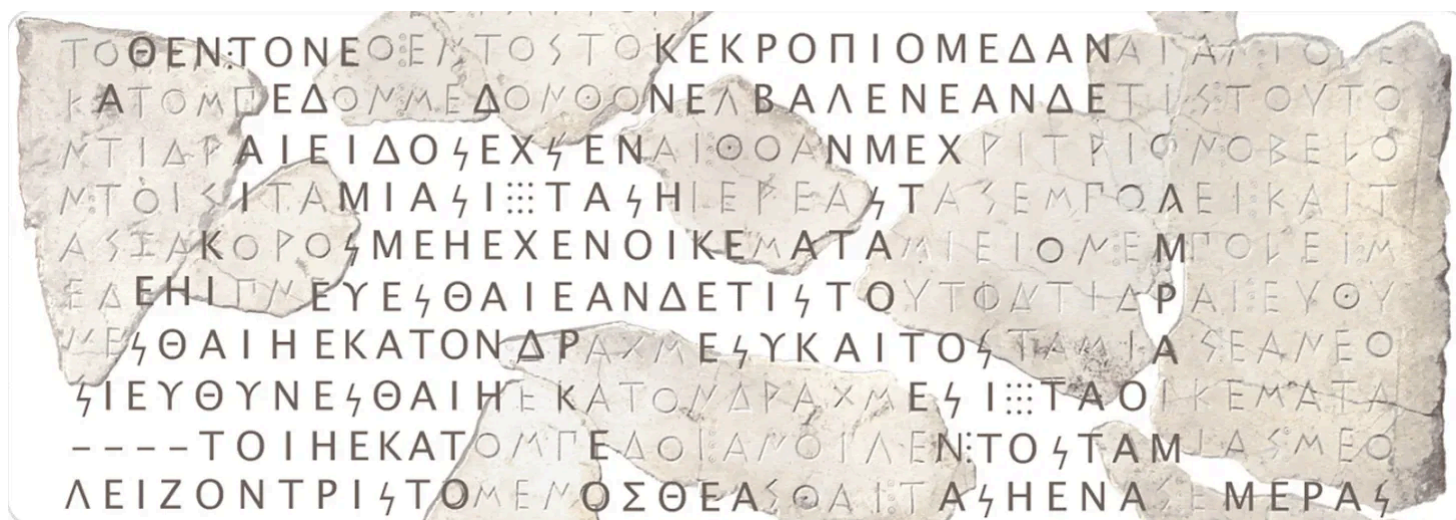
Many of the hardest scientific problems require progress at the boundaries between fields. However when practitioners are brought together, [for example during Covid-19](#), they often struggle to transition from *multidisciplinarity* - where they each retain their own disciplinary angle - to genuine *interdisciplinarity*, where they collectively develop [shared ideas and methods](#). This challenge reflects the growing specialisation of scientific knowledge, as well as [incentives](#) such as grant funding, that often evaluate practitioners predominantly on their core expertise.

许多最困难的科学问题需要在学科交叉的边界上取得进展。然而，当从业者被聚集在一起时，例如在新冠疫情期间，他们往往难以从多学科性——即他们各自保持自己的学科视角——过渡到真正的跨学科性，即在集体中共同发展共享思想和方法的跨学科性。这一挑战反映了科学知识的日益专门化，以及诸如资助资金等激励措施，这些措施通常主要根据从业者的核心专业知识来评估他们。

AI for Science efforts are often multidisciplinary by default, but to succeed, they need to become genuinely interdisciplinary. A starting point is to pick a problem that requires each type of expertise, and then provide enough time and focus to cultivate a team dynamic around it. For example, our [Ithaca](#) project used AI to restore and attribute damaged ancient Greek inscriptions, which could help practitioners to study the thought, language, and history of past civilizations. To succeed, project co-lead Yannis Assael had to [develop](#) an understanding of *epigraphy* - the study of ancient inscribed text. The project's epigraphers, in turn, had to learn how the AI model worked, given the importance of intuition to their work. Cultivating these team dynamics requires the right incentives. Empowering a small, tight-knit team to focus on solving *the problem*, rather than authorship of papers, was key to the AlphaFold 2 breakthrough. This type of focus can be easier to achieve in industry labs, but again highlights the importance of longer-term public research funding that is less tied to

publication pressures.

科学领域的 AI 工作通常默认是多学科的，但要取得成功，它们需要真正实现跨学科。一个起点是选择一个需要各种专业知识的问题，然后提供足够的时间和关注来培养围绕该问题的团队协作。例如，我们的伊萨卡项目使用 AI 修复和归属受损的古希腊铭文，这可以帮助从业者研究过去文明的思想、语言和历史。要取得成功，项目联合负责人 Yannis Assael 不得不了解碑文学——即对古代铭文文本的研究。反过来，项目中的碑文学家们必须学习 AI 模型的工作原理，因为直觉对他们的工作至关重要。培养这些团队协作需要正确的激励机制。赋予一个规模小、关系紧密的团队专注于解决问题，而不是论文的署名权，是 AlphaFold 2 突破的关键。这种专注更容易在工业实验室中实现，但再次强调了长期公共研究资金的重要性，这种资金与发表压力的联系较少。



This image shows a restored decree concerning the Acropolis of Athens and dates 485/5 BCE. (CC BY-SA 3.0, WikiMedia).

这张图片展示了一项关于雅典卫城的重修法令，日期为公元前 485/5 年。(CC BY-SA 3.0, WikiMedia)。

To achieve genuine interdisciplinarity, organisations also need to create roles and career paths for individuals who can help blend disciplines. At Google DeepMind, our *research engineers* encourage a positive feedback loop between research and engineering, while our *programme managers* help to cultivate team dynamics within a research effort and create links across them. We also prioritise hiring individuals who enjoy finding and bridging connections between fields, as well as those that are motivated by rapidly upskilling in new areas. To encourage a cross-pollination of ideas, we also encourage scientists and engineers to regularly switch projects. Ultimately, the goal is to create a culture that encourages curiosity, humility and what the economic historian Joel Mokyr has referred to as '[contestability](#)' - where practitioners of all backgrounds feel empowered to present and constructively critique each other's

work in open talks and discussion threads.

为了实现真正的跨学科合作，组织还需要为能够帮助融合不同学科的个人创建角色和职业发展路径。在 Google DeepMind，我们的研究工程师鼓励研究与工程之间形成积极的反馈循环，而我们的项目管理人员则帮助在研究工作中培养团队协作精神，并建立不同团队之间的联系。我们还优先招聘那些喜欢发现和连接不同领域的人，以及那些在新领域快速提升技能的人。为了促进思想的交流，我们还鼓励科学家和工程师定期更换项目。最终的目标是创建一种鼓励好奇心、谦逊的文化，以及经济学家历史学家乔尔·莫基尔所说的“竞争性”——即所有背景的从业者都能在公开的讨论和讨论串中提出和建设性地批评彼此的工作。

7. ADOPTION: Carefully consider the best access option and spotlight AI models' uncertainties

7. 采用：仔细考虑最佳访问选项，并突出 AI 模型的不确定性

Many AI for Science models, such as AlphaFold or our weather forecasting work, are *specialised* in the sense that they perform a small number of tasks. But they are also *general* in the sense that [a large number of scientists are using them](#), for everything from understanding diseases to improving fishing programmes. This impact is [far from guaranteed](#). The [germ-theory of disease](#) took a long time to diffuse, while the downstream products that scientific breakthroughs could enable, such as [novel antibiotics](#), often lack the right market incentives.

许多 AI for Science 模型，如 AlphaFold 或我们的天气预报工作，在某种程度上是专门化的，因为它们只执行少数任务。但它们在另一方面又是通用的，因为大量科学家在使用它们，从理解疾病到改进渔业项目，用途广泛。这种影响远非必然。病原体致病理论传播了很长时间，而科学突破可能带来的下游产品，如新型抗生素，往往缺乏适当的市场激励。

When deciding how to release our models, we try to balance the desire for widespread adoption and validation from scientists with commercial goals and other considerations, such as potential safety risks. We also created a dedicated Impact Accelerator to drive adoption of breakthroughs and encourage socially beneficial applications that may not otherwise occur, including through partnerships with organisations like the [Drugs for Neglected Diseases Initiative](#), and the [Global Antibiotic](#)

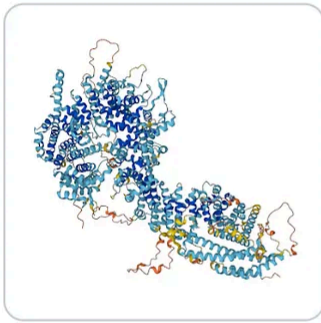
[Research & Development Partnership](#), that have a similar mandate.

在决定如何发布我们的模型时，我们试图在广泛采用和科学家验证的愿望与商业目标和其他考虑因素（如潜在的安全风险）之间取得平衡。我们还创建了一个专门的影响加速器，以推动突破性技术的采用，并鼓励可能不会发生的具有社会效益的应用，包括通过与具有类似使命的组织（如被忽视疾病药物倡议组织和全球抗生素研究与开发伙伴关系）建立合作。

To encourage scientists who could benefit from a new model or dataset to use it, [developers need to make it](#) as easy as possible for scientists to use and integrate into their workflows. With this in mind, for AlphaFold 2 we open-sourced the code but also partnered with [EMBL-EBI](#) to develop [a database](#) where scientists, including those with less computational skills and infrastructure, could search and download from a preexisting set of 200 million protein structures. AlphaFold 3 expanded the model's capabilities, leading to a combinatorial explosion in the number of potential predictions. This created a need for a new interface, the [AlphaFold Server](#), which allows scientists to create structures on-demand. The scientific community has also developed their own AlphaFold tools, such as [ColabFold](#), demonstrating the diversity of needs that exist, as well as the value of nurturing computational skills in the scientific community to address these needs.

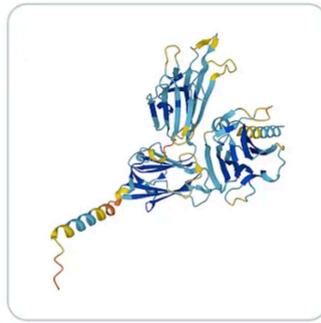
为了鼓励那些能从新模型或数据集中受益的科学家使用它们，开发者需要尽可能简化科学家使用和整合到他们工作流程的过程。基于这一考虑，对于 AlphaFold 2，我们开源了代码，并与 EMBL-EBI 合作开发了一个数据库，科学家们（包括那些计算技能和基础设施较弱的科学家）可以在其中搜索和下载已有的 2 亿个蛋白质结构。AlphaFold 3 扩展了模型的功能，导致潜在预测数量呈指数级增长。这催生了对新界面的需求，即 AlphaFold Server，它允许科学家按需创建结构。科学界还开发了他们自己的 AlphaFold 工具，如 ColabFold，这展示了存在的需求多样性，以及培养科学界计算技能以应对这些需求的价值。

AlphaFold predictions referenced in publications



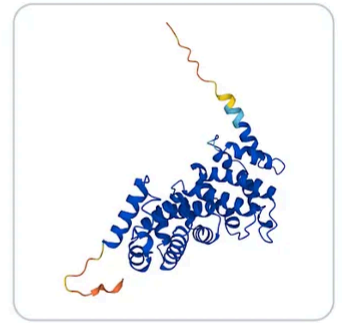
**Nuclear pore complex
protein Nup205**

Part of a large complex that acts as a gateway in and out of the cell nucleus



**Gametocyte surface
protein P45/48**

From the malaria parasite; a candidate protein for including in vaccines



**CCR4-NOT transcription
complex subunit 9**

Regulates an important cellular process (the rate of mRNA degradation)



Ice nucleation protein

Bacterial protein that can trigger ice formation at relatively high temperatures, causing frost damage to plants



F20H23.2 protein

Plant protein; represents a potential new structural superfamily unlike anything seen before



Vitellogenin

Involved in the immune system of egg-laying animals including honeybees

To date, more than 2M users from over 190 countries have accessed the AlphaFold Protein Structure Database to view over 7 million structures.

迄今为止，来自 190 多个国家的超过 200 万用户已访问 AlphaFold 蛋白质结构数据库，查看了超过 700 万个结构。

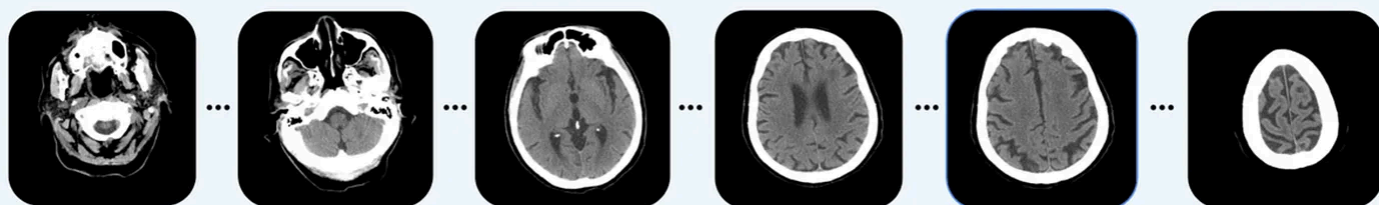
Scientists also need to trust an AI model in order to use it. We expand on the reliability question below, but a useful starting point is to proactively signal how scientists should use a model, as well as its uncertainties. With AlphaFold, following dialogue with scientists the team developed uncertainty metrics that communicated how 'confident' the model was about a given protein structure prediction, supported by intuitive visualisations. We also partnered with EMBL-EBI to develop a [training module](#) that offered guidance on how to best use AlphaFold, including how to interpret the

confidence metrics, supported by practical examples of how other scientists were using it. Similarly, our [Med-Gemini](#) system recently achieved state-of-the-art performance on answering health-related questions. It uses an *uncertainty-guided* approach that responds to a question by generating multiple 'reasoning chains' for how it might answer. It then uses the relative divergence between these initial answers to calculate how uncertain the answer is. Where uncertainty is high, it invokes web search to integrate the latest, up-to-date information.

科学家也需要信任 AI 模型才能使用它。我们将在下文详细探讨可靠性问题，但一个有用的起点是主动向科学家传达如何使用模型，以及其不确定性。以 AlphaFold 为例，在与科学家进行对话后，团队开发了不确定性指标，通过直观的可视化方式传达模型对特定蛋白质结构预测的“置信度”。我们还与 EMBL-EBI 合作开发了一个培训模块，提供如何最佳使用 AlphaFold 的指导，包括如何解读置信度指标，并辅以其他科学家使用该模型的实际案例。类似地，我们的 Med-Gemini 系统最近在回答与健康相关的问题上达到了顶尖性能。它采用不确定性引导方法，通过生成多个“推理链”来回应问题，说明它可能如何回答。然后，它利用这些初始答案之间的相对差异来计算答案的不确定性。当不确定性较高时，它会调用网络搜索来整合最新、最及时的信息。

Prompt

Given the image, what are the radiology findings?



Response

FINDINGS: mild cortical atrophy and periventricular white matter changes are noted compatible with microvasculopathy. there is proportionate ventriculomegaly. no bleed, mass effect, midline shift or abnormal intracranial fluid is seen. no evidence of acute intracranial hemorrhage, territorial infarct, abnormal fluid or mass effect. no abnormal extra-axial fluid collection is identified. no definitive evidence of intracranial hemorrhage or large territorial infarct within the limits of ct imaging. **old left basilar ganglia infarct**. no intracranial mass, swelling, hemorrhage, infarction or extra-axial collection is seen. visualized mastoid air cells: well pneumatized. visualized paranasal sinuses: well pneumatized. atherosclerotic calcification of vasculature at the skull base.

Radiologist Written Report

FINDINGS: the fourth ventricle is midline. the remaining ventricular system is within normal limits. there is mild chronic white matter microangiopathic disease and moderate generalized cortical atrophy. no bleed, mass effect, midline shift or abnormal intracranial fluid is seen. bone windows demonstrate an intact skull. the remaining visualized paranasal sinuses and mastoids are clear. chronic white matter microangiopathic disease, mild generalized cortical atrophy, not significantly changed since 2015. no evidence of acute intracranial hemorrhage, territorial infarct, abnormal fluid or mass effect.

Med-Gemini-3D is able to generate reports for CT scans, a far more complex form of imaging than standard X-rays. In this example, Med-Gemini-3D's report has correctly included a pathology (highlighted in green) that was missed in the original radiologist's report. Note that 'basilar' is a common mis-transcription of 'basal' that Med-Gemini has learned from the training data, though the meaning

of the report is unchanged.

Med-Gemini-3D 能够生成 CT 扫描报告，这是一种比标准 X 光更复杂的成像形式。在这个例子中，Med-Gemini-3D 的报告正确地包含了原始放射科医生报告中遗漏的病理学信息（绿色高亮显示）。请注意，“basilar”是“basal”的一个常见错误转录，Med-Gemini 从训练数据中学习到这一点，尽管报告的含义并未改变。

8. PARTNERSHIPS: Aim for early alignment and a clear value exchange

8. 合作伙伴关系：力求早期协调和明确的价值观交换

AI for Science efforts require a diversity of expertise that creates a strong need for partnership - both formal and informal - between public and private organisations. For example, our [FunSearch](#) method developed a new construction for the [Cap Set](#) problem, which renowned mathematician Terence Tao once described as his [favourite open question](#). This was enabled by collaborating with Jordan Ellenberg, a professor of mathematics at the University of Wisconsin-Madison and a noted Cap Set expert.

科学领域的 AI 工作需要多样化的专业知识，这促使公共和私营组织之间建立正式和非正式的合作伙伴关系。例如，我们开发的 FunSearch 方法为 Cap Set 问题提出了一种新的解决方案，该问题曾被誉为著名数学家陶哲轩最喜欢的开放性问题。这一成果得益于与威斯康星大学麦迪逊分校数学教授、Cap Set 问题专家乔丹·埃伦伯格的合作。

Partnerships are needed throughout the project lifecycle, from creating datasets to sharing the research. In particular, AI labs often need scientists to help evaluate an AI model's outputs. For example, our protein design team partnered with research groups from the [Francis Crick Institute](#) to run wet lab experiments to test if our AI-designed proteins bound to their target and if this had the desired function, such as preventing SARS-CoV-2 from infecting cells. Given the central role played by industry labs in advancing AI capabilities, and the need for rich domain expertise, these kinds of public private partnerships will likely become increasingly important to advancing the AI for Science frontier and may require greater investment, such as more funding to support partnerships teams in universities and public research institutions.

在整个项目生命周期中，从创建数据集到共享研究成果，都需要合作。特别是，人工智能实验室通常需要科学家来帮助评估人工智能模型的输出。例如，我们的蛋白质设计团队与弗朗西斯·克里克研究所的研究小组合作，进行湿实验来测试我们人工智能设计的蛋白质是否与目标结合，以及这种结合是否具有预期功能，例如防止 SARS-CoV-2 感染细胞。鉴于产业实验室在提升人工智能能力中发挥的核心作用，以及丰富领域专业知识的需要，这类公私合作可能会越来越重要，以推动人工智能科学前沿的发展，并可能需要更大的投资，例如为大学和公共研究机构中的合作团队提供更多资金支持。

Developing partnerships is difficult. When starting discussions, it is important to align early on the overall goal and address potentially thorny questions, such as what rights each party should have over the outputs, whether there should be a publication, whether the model or dataset should be open sourced, and what type of licensing should apply. Differences of opinion are natural and often reflect the incentives facing public and private organisations, which in turn vary greatly, depending on factors such as the maturity of the research or its commercial potential. The most successful partnerships involve a clear value-exchange that draws on the strengths of each organisation. For example, more than 2 million users from over 190 countries have used the [AlphaFold Protein Structure Database](#). This required a close collaboration to pair our AI model with the biocuration expertise and scientific networks of [EMBL-EBI](#).

建立合作关系很困难。在开始讨论时，重要的是尽早就总体目标达成一致，并解决可能棘手的问题，例如各方可对成果拥有哪些权利、是否需要发表论文、模型或数据集是否应该开源，以及应适用哪种许可证。意见分歧是自然的，通常反映了公共组织和私营组织面临的激励差异，而这些差异又因研究成熟度或其商业潜力等因素而差异很大。最成功的合作涉及一个能够发挥各组织优势的明确价值交换。例如，来自 190 多个国家的超过 200 万用户已经使用了 AlphaFold 蛋白质结构数据库。这需要紧密合作，将我们的 AI 模型与 EMBL-EBI 的生物信息学专业知识和科学网络相结合。

9. SAFETY & RESPONSIBILITY: Use assessments to explore trade-offs and inspire new types of evaluation methods

9. 安全与责任：利用评估探索权衡并启发新的评价方法

Scientists often disagree, sometimes strongly, about the potential benefits and risks that AI models may have on science, and on wider society. Conducting an ethics and safety assessment can help to frame the discussion and enable scientists to decide whether, and how, to develop a given AI model. A starting point is to identify the most important domains of impact, and to specify these domains at the right level of abstraction. There are increasingly sophisticated frameworks to [identify and categorise different AI risks](#), such as enabling mis- and disinformation. But these frameworks rarely consider the potential benefits of AI in the same domain, such as improving access to high-quality information synthesis, or the trade-offs that can occur, for example if you restrict access to an AI model or limit its capabilities.

Assessments should also clarify their timescales, the relative certainty of any impact, and the relative importance, or *additionality*, of AI, to achieving it. For example, those worried about AI and climate change often focus on the immediate power needed to

train large AI models, while AI proponents often focus on the less immediate, less clear, but potentially much larger downstream benefits to the climate from future AI applications. In carrying out their assessment, AI practitioners should also avoid over-indexing on the model's capabilities, which they will be closer to, and better understand the extent to which third parties will actually use it or be affected by it, which typically requires input from external experts to do well.

科学家们经常对人工智能模型可能对科学和更广泛社会带来的潜在益处和风险存在分歧，有时分歧很大。进行伦理与安全评估有助于界定讨论范围，并使科学家能够决定是否以及如何开发特定的人工智能模型。第一步是识别影响最关键的领域，并在适当的抽象层次上明确这些领域。现在有日益复杂的框架用于识别和分类不同的人工智能风险，例如助长错误信息和虚假信息。但这些框架很少考虑同一领域内人工智能的潜在益处，例如提高高质量信息综合的获取途径，或可能出现的权衡，例如如果你限制对人工智能模型的访问或限制其功能。评估还应明确其时间范围、任何影响的相关确定性，以及人工智能在实现目标方面的相对重要性或附加价值。例如，那些担心人工智能和气候变化的人往往关注训练大型人工智能模型所需的即时能力，而人工智能支持者则往往关注未来人工智能应用对气候带来的不那么直接、不那么明确但潜在可能更大的下游效益。在执行他们的评估时，人工智能从业者还应该避免过度关注模型的性能，因为这将更接近他们，并更好地理解第三方实际上会使用它或受其影响的程度，这通常需要外部专家的输入才能做好。

Practitioners also need new methods to better evaluate the potential risks and benefits of using AI in science. At present, many AI safety evaluations rely on specifying the types of content that a model should not output, and quantifying the extent to which the model adheres to this policy. These evaluations are useful for certain risks posed by using AI in science, such as generating inaccurate content. But for other risks, such as to biosecurity, the idea that we can reliably specify certain types of scientific knowledge as dangerous in advance has been [challenged](#), because of the dual-use nature of scientific knowledge, but also because such efforts tend to focus on what has caused harm historically, such as viruses from past outbreaks, rather than novel risks.

从业者还需要新的方法来更好地评估在科学中使用人工智能的潜在风险和收益。目前，许多人工智能安全评估依赖于指定模型不应输出的内容类型，并量化模型遵守该政策的程度。这些评估对于使用人工智能在科学中带来的某些风险是有用的，例如生成不准确的内容。但对于其他风险，如生物安全风险，我们能够提前可靠地指定某些科学知识为危险的想法受到了挑战，这既因为科学知识的双重用途，也因为这些努力往往集中于历史上造成伤害的原因，例如过去的疫情病毒，而不是新型风险。

A better approach may be to evaluate the [dangerous capabilities](#) of AI models, or the degree to which AI models provide an uplift to humans' dangerous capabilities. In many cases, these capabilities will also be dual-use, such as the ability to help design or execute experimental protocols. The degree to which these AI capabilities point to a risk, or an opportunity, will depend on how potential threat actors are assessed and how access to the model is governed. Beyond safety, evaluating other risks from using AI in science, such as to scientific creativity or reliability (which we discuss below), will require entirely new evaluation methods. Given the difficulty of researching and executing such evaluations, it makes sense to pursue them at the community-level, rather than each lab pursuing siloed efforts.

一个更好的方法可能是评估 AI 模型的危险能力，或者评估 AI 模型对人类危险能力的提升程度。在许多情况下，这些能力也将是双用途的，例如帮助设计或执行实验方案的能力。这些 AI 能力指向风险还是机遇的程度，将取决于对潜在威胁行为者的评估方式以及模型访问权限的管理方式。除了安全之外，评估在科学中使用 AI 带来的其他风险，例如对科学创造力或可靠性的影响（我们将在下文讨论），将需要全新的评估方法。鉴于研究和执行此类评估的困难，在社区层面推进这些评估比每个实验室各自为政更有意义。

Part C: The risks C 部分：风险

[Policy papers](#), [government documents](#) and [surveys of scientists](#) regularly cite certain risks from the growing use of AI in science. Three of these risks - to scientific creativity, reliability, and understanding - mainly relate to how science is practised. Two other risks - to equity and the environment - mainly relate to how science represents and affects wider society. The use of AI is often presented exclusively as a risk to these domains, and the domains, such as scientific reliability, or the environment, are often portrayed in stable, somewhat idealised terms, that can overlook the wider challenges that they face.

政策文件、政府文件和科学家调查报告经常提及 AI 在科学中日益增长的使用所带来的某些风险。其中三种风险——对科学创造力、可靠性和理解力的风险——主要与科学实践方式有关。另外两种风险——对公平性和环境的风险——主要与科学如何代表和影响更广泛的社会有关。AI 的使用通常被专门呈现为对这些领域的风险，而这些领域，如科学可靠性或环境，往往被描绘成稳定、略带理想化的状态，从而可能忽视它们所面临的更广泛挑战。

We believe that using AI in science will ultimately benefit each of these five domains, because there are opportunities to mitigate the risks that AI poses, and to use AI to help address wider challenges in these areas, in some cases profoundly. Achieving a beneficial outcome will likely be harder for inequity, which is ingrained into AI and science at multiple levels, from the make-up of the workforce to the data underpinning research, and for scientific creativity, which is highly subjective and so individuals may reasonably disagree about whether a certain outcome is positive. These nuances increase the value of scientists, policymakers and others articulating their expectations for how using AI in science will affect each of these 5 areas.

我们相信，在科学中使用人工智能最终将惠及这五个领域中的每一个，因为存在减轻人工智能所带来风险的机会，并且可以利用人工智能来帮助解决这些领域更广泛的问题，在某些情况下影响深远。实现有益的结果对于不平等来说可能更困难，因为不平等已经渗透到人工智能和科学的多个层面，从劳动力构成到支撑研究的数据库，以及对于科学创造力来说，它高度主观，因此个人可能合理地不同意某个结果是否积极。这些细微之处增加了科学家、政策制定者和其他人阐述他们对使用人工智能如何影响这五个领域的期望的价值。

5 Risks of using AI to accelerate science



1. Creativity



2. Reliability



3. Understanding



4. Equity



5. Environment

1. CREATIVITY: Will AI lead to less novel, counterintuitive, breakthroughs?

1. 创造力：人工智能会带来更少新颖、反直觉、突破性的创新吗？

Scientific creativity describes the creation of something new that is useful. In practice, the extent to which a scientist views a new idea, method, or output as creative typically rests on more subjective factors, such as its perceived simplicity, counterintuitiveness, or beauty. Today, scientific creativity is [undermined](#) by the relative homogeneity of the scientific workforce, which narrows the diversity of ideas. The pressure on researchers to [‘publish or perish’](#) also incentivises [‘crowd-following’ publications on less risky topics](#), rather than the kind of deep work, or bridging of concepts across disciplines, that [often underpins](#) creative breakthroughs. This may explain why [the share](#) of disruptive scientific ideas that cause a field to veer off into a new direction seems to be declining, beyond what may be normally expected, as science expands.

科学创造力指的是创造具有实用价值的新事物。在实践中，科学家将一个新想法、方法或成果视为创造性的程度，通常取决于更多主观因素，例如其被感知的简单性、反直觉性或美感。如今，科学创造力的发挥受到科学工作者相对同质性的影响，这缩小了思想的多样性。研究人员面临的“不发表就淘汰”的压力，也促使他们在风险较低的主题上进行“随大流”的发表，而不是进行那种能够支撑创造性突破的深度工作，或跨学科概念的联系。这或许可以解释为什么在科学不断发展的过程中，那些足以使一个领域转向新方向的颠覆性科学思想的占比似乎在超出正常预期之外下降。

Some scientists worry that using AI may exacerbate these trends, by undermining the more intuitive, unorthodox, and serendipitous approaches of human scientists, such as [Galileo's hypothesis that the earth rotates on its axis](#). This could happen in different ways. One concern is that AI models are trained to [minimise anomalies](#) in their training data, whereas scientists often *amplify* anomalies by following their intuitions about a perplexing data point. Others worry that AI systems are trained to perform specific tasks, and so relying on them will forgo more serendipitous breakthroughs, such as researchers [unexpectedly](#) finding solutions to problems that they weren't studying. At the community level, some worry that if scientists embrace AI *en masse*, it may lead to a [gradual homogenisation](#) of outputs, for example if LLMs produce similar suggestions in response to the queries of different scientists. Or if scientists [over-focus on](#)

[disciplines and problems](#) that are best-suited to AI.

一些科学家担心使用人工智能可能会加剧这些趋势，通过削弱人类科学家更直观、非传统和偶然的探索方法，例如伽利略关于地球绕轴旋转的假设。这种情况可能以不同的方式发生。一个担忧是人工智能模型被训练以最小化其训练数据中的异常，而科学家则通过遵循他们对令人困惑的数据点的直觉来放大异常。其他人担心人工智能系统被训练以执行特定任务，因此依赖它们将错失更多偶然的突破，例如研究人员意外发现他们未研究的问题的解决方案。在社区层面，一些人担心如果科学家大规模拥抱人工智能，可能会导致输出逐渐同质化，例如如果大型语言模型对不同科学家的查询产生相似的建议。或者如果科学家过度专注于最适合人工智能的学科和问题。

Maintaining support for exploratory research and non-AI research could help to mitigate some of these risks. Scientists could also tailor how they use AI so that it boosts rather than detracts from their own creativity, for example [by fine-tuning LLMs to suggest more personalised research ideas](#), or to help scientists better elicit their own ideas, similar to [our early efforts](#) to develop AI tutors that could help students to better reflect on a problem, rather than just outputting answers to questions. AI [could also enable](#) new types of scientific creativity that may be unlikely to otherwise occur. One type of AI creativity is *interpolation* where AI systems identify novel ideas within their training data, particularly where humans' ability to do this is limited, such as efforts to use AI to [detect anomalies in massive datasets from Large Hadron Collider experiments](#). A second type is *extrapolation*, where AI models generalise to more novel solutions outside their training data, such as the famous [move 37](#) that our [AlphaGo system](#) came up with, that stunned human Go experts, or the novel maths proofs and non-obvious constructions that our [AlphaProof and AlphaGeometry 2](#) systems produced. A third type is *invention*, where AI systems come up with an entirely new theory or scientific system, completely removed from its training data, akin to the original development of general relativity, or the creation of complex numbers. AI systems do not currently demonstrate such creativity, but new approaches could potentially unlock this, such as multi-agent systems that are optimised for different goals, like novelty and counterintuitiveness, or AI models that

are trained to generate novel scientific problems in order to inspire novel solutions.

保持对探索性研究和非 AI 研究的支持有助于缓解这些风险。科学家们还可以调整他们使用 AI 的方式，使其促进而非削弱自身的创造力，例如通过微调 LLMs 来提出更个性化的研究想法，或帮助科学家更好地激发自己的想法，类似于我们早期开发 AI 助教以帮助 学生更好地反思问题的努力，而不是仅仅输出问题的答案。AI 还可以实现新的科学创造力类型，这些类型可能原本不太可能发生。一种 AI 创造力是插值，AI 系统在其训练数据中识别出新颖的想法，尤其是在人类能力有限的情况下，例如使用 AI 检测大型强子对撞机实验数据集中的异常。第二种是外推，AI 模型将其训练数据之外推广到更新颖的解决方案，例如我们 AlphaGo 系统提出的著名走法 37，让人类围棋专家感到震惊，或我们 AlphaProof 和 AlphaGeometry 2 系统产生的新颖数学证明和非显而易见的构造。第三种类型是发明，即 AI 系统提出全新的理论或科学体系，完全脱离其训练数据，类似于广义相对论最初的开发，或是复数的创造。目前 AI 系统尚未展现出这种创造力，但新的方法有可能解锁这一点，例如针对不同目标（如新颖性和反直觉性）进行优化的多智能体系统，或是经过训练以生成新颖科学问题从而激发新颖解决方案的 AI 模型。

2. RELIABILITY: Will AI make science less self-correcting?

2. 可靠性：人工智能会降低科学的自我纠错能力吗？

Reliability describes the ability of scientists to depend upon each others' findings, and trust that they are not due to chance or error. Today, a series of [interrelated challenges](#) weaken the reliability of science, including the p-hacking and publication bias which can lead [researchers to underreport negative results](#); [a lack of standardisation](#) in how scientists carry out routine scientific tasks; mistakes, [for example in how scientists use statistical methods](#); [scientific fraud](#); and [challenges with the peer review process, including a lack of qualified peer reviewers](#).

可靠性描述了科学家们能够依赖彼此的研究成果，并相信这些成果并非偶然或错误。如今，一系列相互关联的挑战削弱了科学的可靠性，包括可能导致研究人员低估负面结果的 p-黑客行为和发表偏差；科学家在执行常规科研任务时缺乏标准化；例如科学家在运用统计方法时出现的错误；科研欺诈；以及同行评审过程中的挑战，包括合格同行评审员的缺乏。

Some scientists worry that AI will exacerbate these challenges as some AI research also features [bad practices](#), such as practitioners cherry-picking the evaluations they use to assess their models' performance. AI models, particularly LLMs, are also prone to 'hallucinate' outputs, including scientific citations, that are false or misleading.

Others worry that LLMs may lead to a flood of low-quality papers similar to those that [‘paper mills’ churn out](#). The community is working on mitigations to these problems, including [good practice checklists](#) for researchers to adhere to and different types of AI factuality research, such as training AI models to [ground their outputs to trusted sources](#), or to help [verify the outputs](#) of other AI models.

一些科学家担心 AI 会加剧这些挑战，因为一些 AI 研究也包含不良做法，例如从业者挑选他们用来评估模型性能的评价。AI 模型，尤其是 LLMs，也容易产生虚假或误导性的输出，包括科学引用。其他人担心 LLMs 可能会导致大量低质量论文的泛滥，类似于那些“论文工厂”生产出来的论文。社区正在努力解决这些问题，包括为研究人员提供遵循的良好实践清单，以及不同类型的 AI 事实性研究，例如训练 AI 模型使其输出基于可信来源，或帮助验证其他 AI 模型的输出。

Scientists could also potentially use AI to improve the reliability of the wider research base. For instance, if AI can help to automate aspects of data annotation or experiment design, this could provide much-needed standardisation in these areas. As AI models get better at grounding their outputs to citations, they could also help scientists and policymakers do more [systematic reviews](#) of the evidence base, for example in climate change, where groups like the Intergovernmental Panel on Climate Change are already struggling to keep up with the [inexorable rise in publications](#). Practitioners could also use AI to help detect mistaken or fraudulent images, or misleading scientific claims, as seen in the [recent trial](#) by the Science journals of an AI image analysis tool. More speculatively, AI could potentially help with aspects of peer review, given that some scientists already use LLMs to help critique their own papers, and to help validate the outputs of AI models, [for example in theorem proving](#). However, there are also reasonable concerns about confidentiality, the ability of AI systems to detect truly novel work, and the need for buy-in from scientists given the

consequential role that peer review plays in processes such as grant approvals.

科学家们也可能利用 AI 来提高更广泛研究基础的可信度。例如，如果 AI 能够帮助自动化数据标注或实验设计的某些方面，这将为这些领域提供急需的标准化。随着 AI 模型在将输出与其引用内容相结合方面变得更好，它们也能帮助科学家和政策制定者进行更系统的证据基础审查，例如在气候变化领域，像政府间气候变化专门委员会这样的组织已经难以跟上出版物数量不断增长的步伐。从业者还可以利用 AI 来帮助检测错误的或欺诈性的图像，或误导性的科学声明，正如科学期刊最近试验 AI 图像分析工具时所看到的那样。更具推测性的是，鉴于一些科学家已经使用 LLMs 来帮助他们评审自己的论文，并帮助验证 AI 模型的输出（例如在定理证明方面），AI 有可能在同行评审的某些方面发挥作用。然而，也存在一些合理的担忧，涉及保密性、人工智能系统识别真正新颖工作的能力，以及鉴于同行评审在诸如拨款审批等过程中的重要影响，需要获得科学家的支持。

3. UNDERSTANDING: Will AI lead to useful predictions at the expense of deeper scientific understanding?

3. 理解：人工智能会导致以牺牲更深层次科学理解为代价的有用预测吗？

In a recent [Nature survey](#), scientists cited a reliance on pattern matching at the expense of deeper understanding as the biggest risk from using AI in science. Understanding is not always necessary to discover new scientific phenomena, such as superconductivity, or to develop useful applications, such as drugs. But most scientists view understanding as one of their primary goals, as the deepest form of human knowledge. Concerns about AI undermining scientific understanding include the argument that modern deep learning methods are *atheoretical* and do not incorporate or contribute to theories for the phenomena that they predict. Scientists also worry that AI models are *uninterpretable*, in the sense that they are not based on clear sets of equations and parameters. There is also a concern that any explanation for an AI model's outputs will not be [accessible or useful](#) to scientists. Taken together, AI models may provide useful predictions about the structure of a protein, or the weather, but will they be able to help scientists understand why a protein folds a

certain way, or how atmospheric dynamics lead to weather shifts?

在最近的一期《自然》杂志调查中，科学家们指出，过度依赖模式匹配而忽视深度理解，是使用人工智能进行科学研究时最大的风险。发现新的科学现象，如超导现象，或开发有用的应用，如药物，并不总是需要理解。但大多数科学家将理解视为他们主要目标之一，因为这是人类最深刻的知识形式。人们担心人工智能会破坏科学理解，包括这样一种观点：现代深度学习方法缺乏理论支撑，无法包含或促进它们所预测现象的理论。科学家们还担心，人工智能模型是不可解释的，因为它们不是基于清晰的方程组和参数建立的。此外，人们还担心，对人工智能模型输出的任何解释都不会对科学家们具有可访问性和实用性。综合来看，人工智能模型可能提供关于蛋白质结构或天气的有用预测，但它们能否帮助科学家理解蛋白质为何以特定方式折叠，或大气动力学如何导致天气变化呢？

Concerns about replacing '*real, theoretical science*' with '*low-brow ... computation*' are not new and [were levelled](#) at past techniques, such as the Monte Carlo method. Fields that merge engineering and science, such as [synthetic biology](#), have also faced accusations of prioritising useful applications over deeper scientific understanding. Those methods and technologies led to gains in scientific understanding and we are confident that AI will too, even if some of these gains will be hard to predict in advance. First, [most AI models are not atheoretical](#) but build on prior knowledge in different ways, such as in the construction of their datasets and evaluations. Some AI models also have interpretable outputs. For example, our [FunSearch](#) method outputs computer code that also describes how it arrived at its solution.

担心用“低俗的……计算”取代“真实的、理论科学”并非新问题，过去的技术，如蒙特卡洛方法，也曾受到类似的指责。将工程与科学相结合的领域，如合成生物学，也面临着优先考虑实用应用而非深入科学理解的指控。这些方法和技术促进了科学理解的进步，我们相信人工智能也会如此，即使其中一些进步难以提前预测。首先，大多数人工智能模型并非无理论依据，而是以不同方式建立在先验知识之上，例如在其数据集构建和评估过程中。一些人工智能模型还提供可解释的输出。例如，我们的 FunSearch 方法会输出计算机代码，同时描述其得出解决方案的过程。

Researchers are also working on explainability techniques that could shed light on how AI systems work, such as [efforts to identify the 'concepts' that a model learns](#). Many of these explainability techniques have [important limitations](#), but they have already enabled scientists to extract new scientific hypotheses from AI models. For example, transcription factors are proteins that bind to DNA sequences to activate or repress the expression of a nearby gene. One [AI research effort](#) was able to predict the relative

contribution of each base in a DNA sequence to the binding of different transcription factors and to explain this result using concepts familiar to biologists. A bigger opportunity may be to learn entirely new concepts based on how AI systems learn. For example, our researchers recently [demonstrated](#) that our AlphaZero system learned ‘superhuman’ knowledge about playing chess, including unconventional moves and strategies, and used another AI system to extract these concepts and teach them to human chess experts.

研究人员还在研究可解释性技术，这些技术能够揭示人工智能系统的工作原理，例如识别模型学习到的“概念”。许多可解释性技术存在重要局限性，但它们已经使科学家能够从人工智能模型中提取新的科学假设。例如，转录因子是能与 DNA 序列结合以激活或抑制附近基因表达的蛋白质。一项人工智能研究工作能够预测 DNA 序列中每个碱基对不同的转录因子结合的相对贡献，并使用生物学家熟悉的“概念”来解释这一结果。更大的机遇可能是基于人工智能系统学习的方式学习全新的概念。例如，我们研究人员最近证明，我们的 AlphaZero 系统能够学习到“超人类”水平的国际象棋知识，包括非常规的走法和策略，并使用另一个人工智能系统提取这些概念，教给人类国际象棋专家。

Even without explainability techniques, AI will improve scientific understanding simply by opening up new research directions that would otherwise be prohibitive. For example, by unlocking the ability to generate a huge number of synthetic protein structures, AlphaFold enabled scientists to search *across protein structures*, rather than just across protein sequences. [One group](#) used this approach to discover an ancient member of the Cas13 protein family that offers promise for editing RNA, including to help diagnose and treat diseases. This discovery also challenged previous assumptions about how Cas13 evolved. Conversely, [efforts to modify](#) the AlphaFold model architecture to incorporate more prior knowledge led to worse performance. This highlights the trade-off that can occur between accuracy and interpretability, but also how AI systems could advance scientific understanding not *in spite of* their opacity, but *because of* it, as this opacity can stem from their ability to operate in high-dimensional spaces that may be uninterpretable to humans, but necessary to making

scientific breakthroughs.

即使没有可解释性技术，人工智能也会通过开辟原本难以实现的研究方向来提升科学理解。例如，通过解锁生成大量合成蛋白质结构的能力，AlphaFold 使科学家能够跨蛋白质结构进行搜索，而不仅仅是跨蛋白质序列。一个研究团队采用这种方法发现了一种古老的 Cas13 蛋白家族成员，该成员为编辑 RNA 提供了前景，包括帮助诊断和治疗疾病。这一发现也挑战了关于 Cas13 如何进化的先前假设。相反，试图修改 AlphaFold 模型架构以纳入更多先验知识的工作导致性能下降。这突出了准确性和可解释性之间可能存在的权衡，但也说明了人工智能系统如何因其不透明性而推进科学理解——这种不透明性源于其能够在人类难以理解的高维空间中运行，而这些高维空间对于实现科学突破却是必要的。

4. EQUITY: Will AI make science less representative, and useful, to marginalised groups?

4. 公平性：人工智能会使科学对边缘化群体来说不那么具有代表性，也不那么有用吗？

Inequity is starkly visible in the scientific workforce, in the questions they study, in the data and models they develop, and in the benefits and harms that result. These inequities are related and can compound over time. For example, a small number of labs and individuals in higher-income cities account for a [disproportionate share of scientific outputs](#). [Studies to identify genetic variants associated with disease](#) rely heavily on data from European ancestry groups, while the [neglected tropical diseases](#) that disproportionately affect poor countries receive relatively little research funding. In agriculture, crop innovations focus on pests that are most common in high-income countries, and are then [inappropriately used on different pests in lower-income countries](#), hurting yields. Despite positive trends, women [account](#) for just 33% of scientists and have [long been underrepresented in clinical trials, particularly women of colour](#).

科学领域中的不平等现象在科研人员构成、研究课题、数据与模型开发以及最终产生的利益与危害等方面都显而易见。这些不平等相互关联，并可能随时间累积加剧。例如，高收入城市中少数实验室和个人占据了不成比例的科研产出份额。旨在识别与疾病相关的基因变异的研究严重依赖欧洲血统人群的数据，而那些主要影响贫困国家的被忽视的热带疾病却相对缺乏研究资金。在农业领域，作物创新主要针对高收入国家常见的害虫，随后却不当用于低收入国家的不同害虫，导致产量受损。尽管存在积极趋势，女性科学家仅占 33%，且长期以来在临床试验中代表性不足，尤其是有色人种女性。

Observers worry that the growing use of AI in science could exacerbate these inequities. AI and computer science workforces are less representative, in terms of [gender, ethnicity and the location of leading labs](#), than many other scientific disciplines and so AI's growing use could hurt broader representation in science. As a data-driven technology, AI also risks inheriting and entrenching the biases found in scientific datasets.

观察者担心，人工智能在科学中的日益广泛使用可能会加剧这些不平等。人工智能和计算机科学领域的工作队伍在性别、种族以及领先实验室的地理位置方面，代表性不如许多其他科学学科，因此人工智能的广泛使用可能会损害科学的整体代表性。作为一种数据驱动技术，人工智能也面临着继承和固化科学数据集中存在的偏见的风险。

There are also opportunities to use AI to reduce inequities in science, albeit not in lieu of more systemic change. If AI models are provided via low-cost servers or databases, they could make it easier and cheaper for scientists, including those from underrepresented groups, to study traditionally neglected problems, similar to how [releasing more satellite data led to more research from underrepresented communities](#). By ingesting more data, AI models may also be able to learn more universal patterns about the complex systems that scientists study, making these models more robust and less prone to biases. For example, owing to their non-representative data, studies that identify genetic variants associated with disease can pick up *confounding*, rather than *causal* variants. Conversely, some early attempts to train [AI models](#) on larger datasets of protein structures and genetic variants, including data across species, perform better at predicting individuals at the greatest risk for disease, with fewer discrepancies across population groups. Ultimately, however, improving equity will require long-term efforts, such as the [H3Africa](#) initiative in genomics and the [Deep Learning Indaba](#) initiative for AI, that aim to build up scientific

infrastructure, communities, and education where it is most lacking.

此外，也有机会利用人工智能来减少科学领域的公平性问题，尽管这并不能替代更系统性的变革。如果通过低成本服务器或数据库提供人工智能模型，它们可以更轻松、更经济地帮助科学家，包括来自代表性不足的群体，研究传统上被忽视的问题，这类似于释放更多卫星数据促使代表性不足的社群进行更多研究。通过处理更多数据，人工智能模型还可能学习到关于科学家研究的复杂系统的更普遍的模式，使这些模型更加稳健，不易产生偏见。例如，由于数据缺乏代表性，识别与疾病相关的基因变异的研究可能会捕捉到混杂变异，而非因果变异。相反，一些早期尝试在更大规模的蛋白质结构和基因变异数据集上训练人工智能模型（包括跨物种的数据），在预测疾病风险最高的个体方面表现更好，且不同人群群体之间的差异更少。最终，然而，改善公平性将需要长期努力，例如在基因组学领域的 H3Africa 倡议和用于人工智能的深度学习 Indaba 倡议，这些倡议旨在建立科学基础设施、社区和教育，这些方面最缺乏。

5. THE ENVIRONMENT: Will AI hurt or help efforts to achieve NetZero?

5. 环境：人工智能会损害还是会帮助实现净零排放？

Given their desire to understand the natural world, many scientists have long been active in efforts to protect the environment, from providing [early evidence](#) about climate change to developing photovoltaic cells. In recent years, a growing number of scientists have [voiced concerns](#) about the potential impact of AI on the environment and developed [methodologies](#) to try to quantify these impacts. Most concerns focus on the [potential impact](#) of training and using LLMs on greenhouse gas emissions, as well as related concerns, such as about the [water needed to cool data centres](#). One way to think about these effects is the [life cycle approach](#), which captures both direct and indirect effects. Direct effects include the emissions from [building](#) and powering the data centres and devices that AI models are trained and run on. There is no comprehensive estimate for *all* direct emissions from AI. However, a 2021 [estimate](#) suggested that cloud and hyperscale data centres, where many large AI models are

trained and deployed, accounted for just 0.1-0.2% of global emissions.

出于理解自然世界的愿望，许多科学家长期以来一直积极参与环境保护工作，从提供关于气候变化的早期证据到开发光伏电池。近年来，越来越多的科学家对人工智能对环境可能产生的影响表示担忧，并开发了方法来尝试量化这些影响。大多数担忧集中在训练和使用 LLMs 对温室气体排放的潜在影响，以及相关的担忧，例如冷却数据中心所需的水资源。思考这些影响的一种方式生命周期方法，该方法涵盖了直接和间接影响。直接影响包括为训练和运行人工智能模型而建设和供电的数据中心和设备所产生的排放。目前还没有针对人工智能所有直接排放的全面估算。然而，2021 年的估算表明，许多大型人工智能模型被训练和部署的云和超大规模数据中心仅占全球排放的 0.1-0.2%。

As the size of LLMs continues to grow, observers have cautioned that these figures may increase, potentially significantly. However, many users of LLMs, including scientists, will be able to fine-tune them, or use their predictions, at a relatively low compute cost, rather than training them from scratch. Efforts are also underway to make LLMs more efficient, and the history of digital technology suggests that sizable gains are possible, not least due to the commercial pressures to deliver faster and cheaper AI models. In some instances, the emissions from AI models will be lower than other approaches. For example, our internal analysis suggests that determining the structures of a small number (<10) of proteins experimentally uses roughly the same energy as a full training run of AlphaFold 2. These results need to be interpreted carefully, as AI simulations rely on, and inform, physical experiments, rather than substituting for them. But they also show how AI could enable a larger amount of scientific activity at a lower average energy cost.

随着 LLMs 规模的持续增长，观察者已警告这些数字可能会增加，甚至可能大幅增加。然而，许多 LLMs 用户，包括科学家，能够以相对较低的计算成本对它们进行微调，或使用它们的预测，而不是从头开始训练。同时也在努力提高 LLMs 的效率，数字技术的历史表明，显著的增长是可能的，这至少部分得益于商业压力以更快、更便宜的 AI 模型交付。在某些情况下，AI 模型的排放量可能低于其他方法。例如，我们的内部分析表明，通过实验确定少量 (<10) 蛋白质的结构所使用的能量大致与 AlphaFold 2 的一次完整训练运行相当。这些结果需要谨慎解读，因为 AI 模拟依赖于并指导物理实验，而不是替代它们。但它们也展示了 AI 如何能够以更低的平均能量成本支持更大规模的科学活动。

Crucially, the *direct* effects of AI on emissions, whether positive or negative, will likely be minor compared to the *indirect* effects that AI-enabled applications have on emissions. Using AI in science opens up three major opportunities to reduce

[emissions](#). First, progress at the [nexus](#) between AI, maths and computer science could dramatically improve the efficiency of the Internet, from [designing more efficient chips](#) to [finding more efficient algorithms](#) for routine tasks. As a growing share of the economy moves online, this should help to offset emissions across these sectors. AI could accelerate the development and use of renewable energy, for example by designing new materials, such as for batteries or solar panels, by [optimising how the grid operates](#) and how it [integrates renewables](#), and via more transformative but uncertain opportunities like fusion. Finally, the world is [already getting warmer](#) and AI could help to better prepare for extreme weather events. For example, our weather forecasting model recently [correctly predicted](#), seven days in advance, that the deadly [Hurricane Beryl](#) would make 'landfall' in Texas. Non-AI models had originally predicted landfall in Mexico before correcting their prediction to Texas three days before it occurred.

关键在于，AI 对排放的直接效应，无论是正面还是负面，与 AI 赋能应用对排放产生的间接效应相比，很可能微不足道。在科学中使用 AI，为减少排放提供了三大机遇。首先，AI、数学和计算机科学之间的进展可能显著提高互联网的效率，从设计更高效的芯片到为常规任务寻找更高效的算法。随着越来越多的经济活动转移到线上，这应有助于抵消这些领域的排放。例如，AI 可以加速可再生能源的开发和使用，通过设计新型材料（如电池或太阳能电池板）、优化电网运行和可再生能源的整合方式，以及通过更具变革性但不确定的机遇（如核聚变）。最后，世界已经变暖，AI 可以帮助更好地应对极端天气事件。例如，我们的天气预报模型最近提前七天准确预测了致命飓风贝拉将在德克萨斯州“登陆”。非 AI 模型最初预测将在墨西哥登陆，但在发生前三天修正了预测为德克萨斯州。

Part D: The policy response

部分 D：政策回应

Given the importance of scientific progress to almost every major economic, environmental and security goal, it follows that science, and the potential for AI to accelerate it, [should be a top priority for any government](#). What should a new AI for Science policy agenda look like? Policymakers can start by implementing the many good science and innovation policy ideas that [already exist](#) and which make even more sense in an era of AI-enabled science. For example, AI will improve the return on science research funding and so it provides a strong rationale to invest more in it and to trial new ideas to [speed up](#) and [experiment with](#) how this funding is allocated. On

compute, governments could implement the idea set out in the UK's [Independent Review](#) to empower a dedicated body to continually assess and advise governments on potential investments. To support AI for Science startups, policymakers can improve their [spin-out policies](#) and support well-run [start-up incubators](#) and [fellowships](#).

鉴于科学进步对几乎所有主要经济、环境和安全目标都至关重要，因此科学——以及 AI 加速科学发展的潜力——理应成为任何政府的首要优先事项。一项新的 AI 科学政策议程应该是什么样子？政策制定者可以从实施许多现有的良好科学和创新政策理念开始，这些理念在 AI 赋能科学的时代更有意义。例如，AI 将提高科学研究的投资回报率，因此有充分的理由增加投资，并试验新想法，以加快和实验如何分配这些资金。在计算方面，政府可以实施英国独立审查中提出的想法，授权专门机构持续评估并向政府提供潜在投资建议。为了支持 AI 科学初创企业，政策制定者可以改进其衍生政策，并支持运营良好的初创企业孵化器和奖学金。

But ambitious new policies are also needed to capitalise on the AI for Science opportunity. We share **four ideas** below. They are intended to be widely applicable, though the precise details would need to be tailored to the specific context of a country, taking into account national priorities, unique strengths and the institutional landscape.

但我们也需要制定雄心勃勃的新政策来把握 AI 为科学带来的机遇。我们在此提出四个想法。这些想法旨在具有广泛的适用性，但具体的细节需要根据各国的具体情况进行调整，同时考虑国家优先事项、独特优势以及制度环境。

1. Define the 'Hilbert Problems' for AI in Science

1. 为科学中的 AI 定义“希尔伯特问题”

Scientific progress rests on picking the right problems. In 1900, the German mathematician David Hilbert [published 23 unsolved problems](#) that proved hugely influential for the subsequent direction of 20th century mathematics. As part of upcoming international events such as the [AI Action Summit](#) in Paris, policymakers, AI labs and science funders could launch a public call for scientists and technologists to identify the most important AI-shaped scientific problems, backed by a major new global fund to drive progress on them. Submissions should specify why the problem is important, why it is suited to modern AI systems, why it may be otherwise neglected, the data bottlenecks that exist, and how near-term technical progress could be

evaluated.

科学进步依赖于选择正确的问题。1900 年，德国数学家大卫·希尔伯特发表了 23 个未解决的问题，这对 20 世纪数学后续的发展产生了巨大影响。在即将到来的国际活动（如巴黎 AI 行动峰会）的框架下，政策制定者、AI 实验室和科学资助者可以发起一项公开征集活动，邀请科学家和技术专家确定最重要的人工智能塑造的科学问题，并设立一项新的重大全球基金来推动这些问题的解决。提交的内容应明确说明该问题的重要性、为什么它适合现代 AI 系统、为什么可能被忽视、现有的数据瓶颈以及如何评估短期内的技术进步。

The top ideas could form the basis of new scientific competitions, where scientists compete to solve these problems with AI, supported by new datasets, evaluation methods and competitive benchmarks. These could build on the [recent flurry of competitions that have emerged to evaluate the scientific capabilities of AI models](#), and include a new [AI for Science Olympiad](#) to attract exceptional young talent from across the world to the field. Beyond its direct impacts, the AI for Science 'Hilbert Problems' initiative could provide a welcome focal point for international scientific collaboration and funding, and inspire a new generation of interdisciplinary scientists to identify and pursue AI-shaped problems.

顶尖想法可以成为新科学竞赛的基础，科学家们将利用人工智能解决这些问题，并得到新数据集、评估方法和竞争基准的支持。这些竞赛可以建立在近期涌现的大量评估人工智能模型科学能力的竞赛基础上，并包括一个新的科学奥林匹克人工智能竞赛，吸引世界各地的杰出青年人才加入该领域。除了直接的影响外，人工智能科学“希尔伯特问题”倡议可以提供一个受欢迎的国际科学合作和资金焦点，并激励新一代跨学科科学家识别和追求人工智能塑造的问题。

2. Make the world readable to scientists

2. 让世界对科学家来说可读

Most scientific data is uncollected, partial, uncurated or inaccessible, making it unavailable to train AI models. There is no single policy response to what is far from a uniform challenge. Policymakers and funders will need to blend a small number of top-down initiatives with support to scale up [promising grassroots efforts](#). A new international network of *AI for Science Data Observatories* should be set up to help address these goals. These Observatories could be given long-term backing and tasked with [running](#) rapid AI for Science 'data stocktakes', where expert teams map the state of data in priority disciplines and application areas. Stocktakes could identify

existing datasets, such as the [Sequence Read Archive](#), whose quality could be further improved, as well as untapped or underutilised datasets, such as the decades of experimental fusion data that is currently unavailable to scientists or leading biodiversity datasets that are subject to restrictive licensing conditions. The stocktakes could also include new 'data wish lists'. For example, our internal analysis suggests that less than 7% of papers in key environmental research domains use AI. We recently funded Climate Change AI to [identify datasets](#) which, if available or improved, could remove some of the bottlenecks to higher AI use. To ensure this analysis leads to action, policymakers should designate and empower organisations to be accountable for addressing the results of the data stocktakes.

大多数科学数据未被收集、不完整、未经整理或无法获取，导致其无法用于训练人工智能模型。对于这一远非统一的挑战，没有单一的应对政策。政策制定者和资助者需要将少数自上而下的倡议与支持扩大有前景的草根努力相结合。应建立一个新的国际人工智能科学数据观测网络，以帮助实现这些目标。这些观测站可以获得长期支持，并被赋予运行快速人工智能科学“数据盘点”的任务，由专家团队绘制优先学科和应用领域的数据现状。盘点可以识别现有数据集，例如序列读取存档，其质量可以进一步改进，以及尚未开发或利用不足的数据集，例如目前科学家无法获取的数十年的实验聚变数据，或受限制许可条件约束的领先生物多样性数据集。盘点还可以包括新的“数据愿望清单”。例如，我们的内部分析表明，在关键环境研究领域，不到 7% 的论文使用人工智能。我们最近资助了 Climate Change AI，以识别那些如果可用或得到改进，就能消除一些阻碍更高 AI 使用的瓶颈的数据集。为确保这项分析能促成行动，政策制定者应指定并授权组织对数据盘点结果负责。

The observatories could also scope the creation of new databases, including ensuring that adequate consideration is given to their long-term storage, maintenance, and incentives. This could include new databases to securely store the results of strategic wet lab experiments that are currently discarded, complemented by making the deposition of these experimental results a requirement for public research funding. Or digitising more public archives, following the example of a recent [UK government and Natural History Museum collaboration](#) to digitise their natural science collections, which includes more than 137 million items, from butterflies to legumes, across a 4.6 billion-year history. Policymakers can also empower scientists to use LLMs to create and improve their own datasets, by ensuring that publicly-funded research is open by default, where possible, building on recent examples from the [UK](#), [US](#) and [Japan](#), including mandates [to release research via pre-print servers](#). Policymakers could seek

to co-fund the most ambitious dataset initiatives with industry and philanthropy.

天文台还可以规划新数据库的创建，包括确保充分考虑其长期存储、维护和激励措施。这可能包括建立新数据库来安全存储目前被丢弃的战略湿实验结果，并通过将提交这些实验结果作为公共研究资助的必要条件来补充。或者，可以借鉴英国政府和自然历史博物馆最近合作数字化其自然科学收藏的例子，这些收藏包括超过 1.37 亿件物品，从蝴蝶到豆科植物，跨越 46 亿年的历史。政策制定者还可以通过确保公共资助的研究在可能的情况下默认开放，从而赋予科学家使用 LLMs 创建和改进他们自己的数据集的权力，借鉴英国、美国和日本最近的例子，包括强制通过预印本服务器发布研究成果。政策制定者可以寻求与产业和慈善机构共同资助最雄心勃勃的数据集计划。

3. Teach AI as the next scientific instrument

3. 将 AI 作为下一代科学仪器进行教学

Over the past half century, as the number of scientific technologies has grown, so has most scientists' distance from them. Many technologies are the products of science, but [an ever smaller share of scientists](#) are trained in how to develop and use them effectively. The pressing near-term need is to fund and incentivise mass uptake of shorter, more tactical AI training programmes [and fellowships](#), for existing scientists and research leaders. Policymakers can incentivise these efforts by setting a clear goal that every postgraduate science student should be able to access introductory courses on using AI in science, including on the most important tools in their domain, in the same way as basic statistics is often taught today. The type and depth of training needed will depend on an individual's discipline and profile, and could range from basic introductory courses about how to reliably use LLMs for everyday research tasks, through to more advanced courses on how to fine-tune AI models on scientific data, as well as how to address more complex challenges, such as evaluating whether the data they used to test a model's performance has intentionally or unintentionally 'leaked' into the data used to train it. These programmes could build on established examples such as the University of Cambridge's [Accelerate Programme](#) that provides structured training in AI to PhD and postdoctoral researchers, or the [short courses](#) that The Carpentries offer on the programming, data, and computational skills needed

to do research.

在过去半个世纪里，随着科学技术数量的增长，大多数科学家与这些技术的距离也在拉大。许多技术是科学的产品，但越来越少比例的科学家接受过如何有效开发和使用它们的培训。迫在眉睫的近期需求是资助和激励大规模采用更短、更具战术性的 AI 培训项目和奖学金，为现有科学家和研究领导者提供支持。政策制定者可以通过设定一个明确的目标来激励这些努力，即每位研究生科学学生都应该能够像今天通常教授基础统计学一样，接触到关于在科学中使用 AI 的入门课程，包括他们领域内最重要的工具。所需培训的类型和深度将取决于个人的学科背景和职业特点，可能从关于如何可靠地使用 LLMs 进行日常研究任务的基础入门课程，到更高级的课程，如如何在科学数据上微调 AI 模型，以及如何应对更复杂的挑战，例如评估用于测试模型性能的数据是否有意或无意地“泄露”到用于训练模型的数据中。这些项目可以借鉴现有的成功案例，例如剑桥大学的加速计划，该计划为博士和博士后研究人员提供结构化的 AI 培训，或者 The Carpentries 提供的关于进行研究所需的编程、数据计算技能的短期课程。

Policymakers also need to move quickly to put in place longer-term programmes to ensure that the next generation of scientists has the skills they need. This means mainstreaming and deepening AI training and skills development in science education at all levels. Secondary school science students will need [early exposure](#) to the impact of AI while university students will need access to [new types of interdisciplinary AI science degrees](#), such as the [pan-African AI for Science Masters programme](#) that we [partnered with](#) the African Institute for Mathematical Sciences to develop. [Dedicated scholarships](#) could also help. For example, the UK's BIG Scholarships programme provides [outstanding opportunities](#) to high school students, with a focus on those from underrepresented groups who have excelled in International Science Olympiads and want to continue their study in leading science hubs but lack the funds to do so.

政策制定者也需要迅速采取行动，建立长期计划，以确保下一代科学家具备所需的技能。这意味着要在各级科学教育中普及和深化人工智能培训与技能发展。中学科学学生需要早期接触人工智能的影响，而大学生则需要能够接触新型跨学科人工智能科学学位，例如我们与非洲数学科学研究所合作开发的全非人工智能科学硕士项目。专门的奖学金也能提供帮助。例如，英国的 BIG 奖学金项目为高中生提供了卓越的机会，重点关注那些来自代表性不足群体、在国际科学奥林匹克竞赛中表现出色并希望继续在领先科学中心学习但缺乏资金的学生。

4. Build evidence and experiment with new ways of organising science

4. 建立证据并尝试以新的方式组织科学

Scientists' use of AI is growing exponentially, but policymakers have little evidence about who is doing it best, how they are doing it, and the hurdles that are inhibiting others. This evidence gap is an impediment to identifying the best AI for Science policy ideas and targeting them effectively. Historically, answers to such questions often come from fields such as economics or innovation studies, but the results can [take years to arrive](#). We are using citation data analysis, interviews, and community engagement to understand how scientists are using our AI models. Governments are [also investing](#) in these *metascience* capabilities to improve how they fund, share and evaluate science research. Building on this momentum, scientists could be tasked with a mission to rapidly assess foundational policy questions, including: where is the most impactful AI for Science research occurring and what types of organisations, talent, datasets, and evaluations are enabling it? To what extent are scientists using and fine-tuning LLMs vs more specialised AI models, and how are they accessing these models? To what extent is AI *actually* benefiting or harming scientific creativity, reliability, the environment, or other domains? How is AI affecting a scientist's perception of their job and what skills, knowledge gaps, or other barriers are preventing their broader use of AI?

科学家对 AI 的应用正呈指数级增长，但政策制定者缺乏关于谁做得最好、如何做到以及哪些障碍阻碍他人的证据。这一证据空白阻碍了识别最适合科学政策的 AI，并有效针对这些 AI。历史上，这类问题的答案往往来自经济学或创新研究等领域，但结果可能需要数年才能得出。我们正在通过引文数据分析、访谈和社区参与来了解科学家如何使用我们的 AI 模型。政府也在投资这些元科学研究能力，以改进其资助、分享和评估科学研究的机制。基于这一势头，科学家可以被赋予任务，快速评估基础政策问题，包括：最具影响力的科学研究的 AI 发生在哪里，以及哪些类型的组织、人才、数据集和评估在推动其发展？科学家在多大程度上使用和微调 LLMs 与更专业的 AI 模型，以及他们如何获取这些模型？AI 在多大程度上真正有益或有害于科学创造力、可靠性、环境或其他领域？AI 如何影响科学家对其工作的认知，以及哪些技能、知识差距或其他障碍阻碍了他们更广泛地使用 AI？

Beyond informing robust policy responses, this evidence base will arm policymakers with the foresight they need to anticipate how AI will transform science and society, similar to the foresight they are developing for AI safety risks through the growing network of [AI Safety Institutes](#). The evidence will also highlight opportunities to reimagine the incentives and institutions needed for science in the age of AI. In particular, scientists and policymakers have [only explored a small fraction](#) of the possible approaches to organising and executing science research. The rise of AI

provides a welcome forcing function to experiment with new types of institutions, from those with more freedom to pursue [high-risk, high-reward](#) research, to Focused Research Organisations aimed at addressing specific bottlenecks. And from new types of [interdisciplinary AI science institutes](#) in priority domains such as climate or food security, to completely novel institutions that we are yet to imagine. Those who experiment faster will stand to benefit the most from a new golden age of discovery.

除了为制定稳健的政策响应提供信息外，这一证据基础还将为政策制定者提供他们所需的远见，以预见人工智能将如何改变科学和社会，类似于他们通过日益增长的 AI 安全研究所网络正在发展的对 AI 安全风险的远见。这些证据还将突出重塑人工智能时代科学研究所需激励和机构的机遇。特别是，科学家和政策制定者仅探索了组织和管理科学研究可能方法中的一小部分。人工智能的兴起提供了一个受欢迎的推动力，以实验新的机构类型，从那些有更多自由追求高风险、高回报研究的机构，到旨在解决特定瓶颈的聚焦研究组织。从在气候或粮食安全等优先领域建立新型跨学科人工智能科学研究所，到我们尚未想象到的完全新颖的机构。那些实验速度更快的人将从新的发现黄金时代中获益最多。

Acknowledgements 致谢

Thank you to Louisa Bartolo, Zoë Brammer and Nick Swanson for research support, and to the following individuals who shared insights with us through interviews and/or feedback on the draft. All views, and any mistakes, belong solely to the authors.

感谢 Louisa Bartolo、Zoë Brammer 和 Nick Swanson 提供研究支持，以及通过访谈和/或对草稿的反馈与我们分享见解的以下个人。所有观点以及任何错误均仅属于作者。

Žiga Avsec, Nicklas Lundblad, John Jumper, Matt Clifford, Ben Southwood, Craig Donner, Joëlle Barral, Tom Zahavy, Been Kim, Sebastian Nowozin, Matt Clancy, Matej Balog, Jennifer Beroshi, Nitarshan Rajkumar, Brendan Tracey, Yannis Assael, Massimiliano Ciaramita, Michael Webb, Agnieszka Grabska-Barwinska, Alessandro Pau, Tom Lue, Agata Laydon, Anna Koivuniemi, Abhishek Nagaraj, Tom Westgarth, Guy Ward-Jackson, Harry Law, Arianna Manzini, Stefano Bianchini, Sameer Velankar, Ankur Vora, Sébastien Krier, Joel Z Leibo, Elisa Lai H. Wong, Ben Johnson, David Osimo, Andrea Huber, Dipanjan Das, Ekin Dogus Cubuk, Jacklynn Stott, Kelvin Guu, Kiran Vodrahalli, Sanil Jain, Trieu Trinh, Rebeca Santamaria-Fernandez, Remi Lam, Victor Martin, Neel Nanda, Nenad Tomasev, Obum Ekeke, Uchechi Okereke,

Francesca Pietra, Rishabh Agarwal, Peter Battaglia, Anil Doshi, Yian Yin.

齐加·阿夫塞克, 尼克拉斯·伦德布拉德, 约翰·琼珀, 马特·克利福德, 本·南伍德, 克雷格·唐纳, 乔埃勒·巴拉尔, 汤姆·扎哈维, 比恩·金, 塞巴斯蒂安·诺沃津, 马特·克莱尼, 马特·巴洛格, 詹妮弗·贝罗什, 尼塔尔尚·拉贾库马尔, 布兰登·特雷西, 扬尼斯·阿萨埃尔, 马西米利亚诺·奇亚拉米塔, 迈克尔·韦伯, 阿格涅什卡·格拉布卡-巴尔文斯卡, 亚历山德罗·保, 汤姆·刘, 阿加塔·莱登, 安娜·科伊维尼米, 阿比什克·纳加拉吉, 汤姆·韦斯特加斯, 盖伊·沃德-杰克逊, 哈里·劳, 阿丽亚娜·曼齐尼, 斯特凡诺·比安奇尼, 萨米尔·维兰卡尔, 安库尔·沃拉, 塞巴斯蒂安·克里尔, 乔尔·Z·莱博, 伊莎贝拉·莱·王, 本·约翰逊, 大卫·奥斯 imo, 安德烈亚·胡伯, 迪帕尼·达斯, 埃金·多古斯·库布克, 杰克琳·斯托特, 凯文·古, 基兰·沃德拉哈利, 萨尼尔·贾因, 崔·林, 丽贝卡·桑塔玛利亚-费尔南德斯, 雷米·拉姆, 维克多·马丁, 尼尔·南达, 内纳德·托马塞夫, 奥布姆·埃克凯, 乌切奇·奥克雷克, 弗朗塞斯卡·皮埃特拉, 里希什·阿格拉瓦尔, 彼得·巴塔格利亚, 阿尼尔·多西, 闫岩。

Errata & updates 勘误与更新

Note: In May 2025, we updated this essay to remove a reference to a now discredited [study](#) on the use of AI in material science by Aidan Toner-Rogers. Stuart Buck has a good [write-up](#) of the warning signs about this study, which we unfortunately missed. A good reminder to be appropriately skeptical with big claims!

注意：2025 年 5 月，我们更新了这篇论文，删除了对艾丹·托纳-罗杰斯关于人工智能在材料科学应用方面现已遭驳斥的研究的引用。斯图尔特·巴克对这一研究的警示信号有很好的概述，不幸的是我们错过了。这提醒我们在面对重大声明时保持适当的怀疑态度！

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AI政策的观点订阅了一个新的发现黄金时代，夺取了AI的科学机会 +1AI政策观点， Conor Griffin, Hanna Schieve和其他2个其他Nnov 26, 2024491310 Sharethis sharethis seasay撰写，由Conor Griffin, Don Wallace, Juan Mateos-Garcia, Hanana Schiefe and PulteMeet Kohli撰写。订阅以获取未来的论文并在下面发表评论，以使我们科学机会，成分，风险和公共政策思想的最重要的AI看法。本文最初发布在Google DeepMind网站上。引言世界各地的实验室正在酿造一场安静的革命，科学家对AI的使用呈指数增长。现在，DostDoc中的三分之一使用大型语言模型来帮助进行文献综述，编码和编辑。10月，我们的Alphafold 2系统的创建者Demis Hassabis和John Jumper成为化学领域的诺贝尔奖获得者，因为他使用AI来预测蛋白质的结构，以及科学家David Baker，以设计新蛋白质。社会将很快开始更直接地感受到这些好处，借助AI目前在开发过程中设计的毒品和材料。在本文中，我们参观了AI如何将科学学科从基因组学转变为计算机科学转变为天气预报。一些科学家正在培训自己的AI模型，而另一些科学家正在微调现有的AI模型，或者使用这些模型的预测来加速其研究。这些科学家正在使用AI作为科学工具来帮助解决重要的问题，例如设计蛋白质更紧密地与疾病靶标结合，但也逐渐改变了科学本身的实施方式。科学家对AI的拥抱背后的势力越来越大。近几十年来，科学家们继续取得相应的进步，从199疫苗到可再生能源。但是，要做这些突破并将其转变为下游应用程序需要大量的研究人员。结果，即使科学劳动力在过去的半个世纪中已经显着增长，仅在美国就增加了七倍以上，但我们期望遵循的社会进步也有所放缓。例如，全世界的大部分地区都见证了生产率增长的持续放缓，这会破坏公共服务的质量。朝着2030年可持续发展目标的进展，该目标捕捉了健康，环境及其他地区最大的挑战。特别是，希望取得突破的科学家越来越多地面临与规模和复杂性有关的挑战，从他们掌握的不增长的文献基础到他们想要运行的日益复杂的实验。现代深度学习方法特别适合这些规模和复杂性挑战，可以压缩未来科学进步所需的时间。例如，在结构生物学中，根据蛋白质的不同，单个X射线晶体学实验以确定蛋白质的结构可能需要多年的工作，并且花费大约100,000美元。Alphafold蛋白结构数据库现在可以免费访问2亿个预测的蛋白质结构。AI对科学的潜在好处不能保证。很大一部分科学家已经使用基于LLM的工具来协助日常任务，例如编码和编辑，但是使用以AI为中心的研究方法的科学家所占的份额要低得多，尽管迅速上升。在急于使用AI中，一些早期的科学用例有可疑的影响。决策者可以帮助加速AI的使用，并将其转向更高影响的地区。美国能源部，欧洲委员会，英国皇家学会和美国国家科学院最近都认可了科学机会的AI。但是，尚无国家制定一项全面的策略来实现它。我们希望我们的文章能够为这种策略提供信息。它针对那些制定和影响科学政策和资金决定的人。我们首先确定了5个机会，其中越来越有必要在科学中使用AI并检查在这些领域取得突破所需的主要成分。然后，我们探讨了在科学中使用AI的最常见的风险，例如科学创造力和可靠性，并认为AI最终可以在每个领域都受益。我们以四种公共政策思想结束，以帮助迎来新的AI-NEA-NEA-SOCICAL。在整篇文章中，我们借鉴了来自Google DeepMind和外部专家的AI专家的两次专家的二十多次访谈的见解。这篇文章自然反映了我们作为私营部门实验室的有利位置，但我们认为我们所提出的案例与整个科学有关。我们希望读者能够通过分享对科学机会，成分，风险和公共政策思想最重要的AI的回应。感谢您阅读AI政策观点！订阅免费获得新帖子并支持我们的工作。SubscripartA。科学家的机会旨在理解，预测和影响自然世界和社会世界如何运作，激发和满足好奇心以及解决社会面临的重要问题。技术和方法，例如显微镜，X射线衍射和统计，既是科学的产物，也是IT的推动者。在过去的一个世纪中，科学家越来越依靠这些工具来进行实验并提高理论。计算工具和大规模数据分析变得尤为重要，从发现希格斯玻色子到人类基因组的映射，一切都可以实现。从一个角度来看，科学家对AI的日益增长是这种长期趋势的逻辑扩展。但这也可能表明了一些更深刻的事物 - 科学能力的限制是不连续的飞跃。这些机会跨学科适用，并解决与规模和复杂性有关的特定瓶颈，科学家越来越多地在科学过程中面临

的不同点，从产生强大的新颖假设到与世界分享他们的工作。

1. 知识：改变科学家如何消化和传达知识以做出新的发现，科学家需要掌握一种先前存在的知识，这些知识继续成倍增长并变得越来越专业。这种“知识负担”有助于解释为什么进行变革性发现的科学家越来越老，跨学科，并且位于精英大学，以及为什么个人或小型团队撰写的论文所占的份额正在下降，即使小型团队通常会更好地取得更好的位置以提高破坏性的科学思想。在分享他们的研究方面，诸如预印式服务器和代码存储库之类的欢迎创新，但大多数科学家仍然以密集的，术语繁重的仅英语论文分享他们的发现。这可能会阻碍对科学家的工作，包括政策制定者，企业和公众的兴趣。科学家已经在使用LLM和基于LLM的早期科学助理来帮助解决这些挑战，例如通过综合文献中最相关的见解。在早期的演示中，我们的科学团队使用我们的双子座LLM在一天之内从最相关的200,000篇论文中找到，提取和填充特定数据。即将到来的创新，例如在更科学的数据上进行微调LLM，以及在长篇小说窗口中的进步和引文使用中的进步，将稳步提高这些功能。随着我们在下面的扩展，这些机会并非没有风险。但是它们提供了一个从根本上重新考虑某些科学任务的窗口，例如在一个世界上“阅读”或“写”科学论文的含义，在这个世界上，科学家可以使用LLM来帮助批评它，对其对不同的受众进行启示，或将其转变为“交互式论文”或音频指南。

2. 数据：尽管叙述了有关数据丰度的时代，但生成，提取和注释大型科学数据集，但大多数自然和社会世界的科学数据长期缺乏科学数据，从土壤，深海和气氛到非正式经济。AI可以以不同的方式提供帮助。它可以使现有数据收集更加准确，例如，通过减少测序DNA，检测样品中的细胞类型或捕获动物声音时可能发生的噪声和错误。科学家还可以利用LLM跨图像，视频和音频运行的能力，以提取埋藏在科学出版物，档案馆以及诸如教学视频之类的不太明显资源中的非结构化科学数据，并将其转换为结构化的数据集。AI还可以通过科学家为使用它所需的支持信息来注释科学数据。例如，至少有三分之一的微生物蛋白是可靠地注释的，其中包含有关它们所执行的功能的详细信息。在2022年，我们的研究人员使用AI来预测蛋白质的功能，从而导致Uniprot，PFAM和InterPro数据库中的新条目。一旦经过验证的AI模型也可以用作合成科学数据的新来源。例如，我们的字母蛋白质设计模型对来自AlphaFold 2的超过1亿个AI生成的蛋白质结构进行了训练，以及蛋白质数据库的实验结构。这些AI机会可以补充和增加其他急需的努力来生成科学数据的回报，例如数字化档案，或资助新的数据捕获技术和方法，例如在单细胞基因组学中进行的努力，以空前的细节创建有力的单个细胞数据集。

3. 实验：模拟，加速并告知复杂的实验：昂贵，复杂且缓慢。有些人根本没有发生，因为研究人员无法访问所需的设施，参与者或投入。融合是一个很好的例子。它承诺了一种能源，该能源实际上是无限的，无排放的，并且可以使能源密集型创新的缩放，例如淡化。为了实现融合，科学家需要创建和控制等离子体 - 第四个基本状态。但是，所需的设施非常复杂。ITER的原型Tokamak反应堆于2013年开始构建，但是直到最早的2030年代中期，等离子体实验才开始进行，尽管其他人希望在较短的时间表上构建较小的反应堆。AI可以帮助模拟融合实验，并可以更有效地利用后续实验时间。一种方法是在物理系统的模拟上运行增强学习剂。在2019年至2021年之间，我们的研究人员与瑞士联邦技术学院Lausanne合作，演示了如何使用RL在模拟Tokamak反应堆的模拟中控制等离子体的形状。这些方法可以扩展到其他实验设施，例如粒子加速器，望远镜阵列或重力波检测器。使用AI模拟实验在整个学科中看起来会大不相同，但是一个共同的线程是，模拟通常会为物理实验提供信息和指导，而不是代替它们。例如，普通人在其DNA中有9,000多个错义变体或单个字母替代。这些遗传变异中的大多数都是良性的，但是有些可以破坏蛋白质的作用，从而导致罕见的遗传疾病，例如囊性纤维化以及诸如癌症等常见疾病。测试这些变体效果的物理实验通常仅限于单个蛋白质。我们的字母敏感模型将7100万个潜在的人类错义变体中的89%分类为有害或良性，使科学家能够将其物理实验集中在最有可能的疾病的贡献者上。alphamissense预测了所有可能的7100万个错义变体的致病性。它分类为89% - 预测57%的可

能是良性，而32%可能是致病性的4。模型：模型复杂系统以及它们的组件如何相互作用，在1960年的论文中，诺贝尔奖获得了诺贝尔奖的物理学家Eugene Wigner，这对模拟重要的自然现象的数学方程式（例如Planetary Motion）的“不合理效力”惊叹于“不合理的有效性”。但是，在过去的半个世纪中，依靠方程组或其他确定性假设集模型一直在努力捕捉生物学，经济学，天气和其他地方系统中系统的全部复杂性。这反映了构成这些系统的大量相互作用零件，以及它们的活力和出现的新兴，随机或混乱的行为的潜力。建模这些系统的挑战阻碍了科学家预测或控制其行为的能力，包括在冲击或干预期间，例如温度升高，新药或引入税收变更。AI可以通过摄入有关它们的更多数据并在此数据中学习更多强大的模式和规律性来更准确地对这些复杂的系统进行建模。例如，现代天气预报是科学和工程学的胜利。对于政府和工业，它可以为从可再生能源计划到为飓风和洪水做准备的所有信息。对于公众来说，天气是Google搜索中最受欢迎的非品牌查询。传统的数字预测方法基于精心定义的物理方程，可提供大气复杂动力学的非常有用但不完美的近似。它们在计算上也很昂贵。在2023年，我们发布了一个深度学习系统，该系统可预测天气情况最多提前10天，这在准确性和预测速度方面优于传统模型。随着我们在下面的扩展，使用AI预测天气变量也可能有助于减轻和应对气候变化。例如，当飞行员飞过潮湿的地区时，可能会导致凝结径，从而导致航空的全球变暖影响。Google科学家最近使用AI来预测何时何地出现潮湿区域，以帮助飞行员避免飞行。在许多情况下，AI会丰富传统的建模复杂系统的方法，而不是取代它们。例如，基于代理的建模模拟了企业和消费者等单个参与者之间的互动，以了解这些相互作用如何影响更大的更复杂的系统，例如经济。传统方法要求科学家事先指定这些计算代理应如何行事。我们的研究团队最近概述了科学家如何使用LLM来创建更灵活的生成代理，以沟通和采取行动，例如搜索信息或进行购买，同时还要推理和记住这些行动。科学家还可以利用强化学习来研究这些代理如何在更动态的模拟中学习和适应其行为，例如回应引入新的能源价格或大流行反应政策。

5。解决方案：确定大型搜索空间的问题的新颖解决方案重要的科学问题，实际上是无法理解的潜在解决方案。例如，生物学家和化学家旨在确定分子（例如蛋白质）的结构，特征和功能。这种工作的目标之一是帮助设计这些分子的新型版本，以用作抗体药物，降解酶或新材料。但是，为了设计一种小分子药物，科学家面临1060多种潜在选择。要设计具有400个标准氨基酸的蛋白质，它们面临20400种选择。这些较大的搜索空间不仅限于分子，而是许多科学问题的司空见惯，例如找到数学问题的最佳证明，计算机科学任务的最有效算法或计算机芯片的最佳体系结构。传统上，科学家依靠直觉，反复试验，迭代或蛮力计算的某种组合来找到最佳的分子，证明或算法。但是，这些方法难以利用潜在的解决方案的巨大空间，使得更好的解决方案未被发现。AI可以打开这些搜索空间的新部分，同时也更快地将最有可能可行和有用的解决方案归为归纳 - 一种微妙的平衡行为。例如，在7月，我们的字母隔热和字母计量计2系统在国际数学奥林匹克（一项精英高中竞赛）中正确解决了六个问题中的四个问题。该系统利用我们的Gemini LLM体系结构为给定数学问题生成了许多新颖的想法和潜在的解决方案，并将其与基于数学逻辑的系统结合使用，这些系统可以在最有可能正确的候选解决方案上迭代地工作。人工智能科学家还是授权科学家？AI在科学中日益增长的使用以及早期AI科学助手出现了人们的疑问，即AI的能力可能进步的速度和程度，这对人类科学家意味着什么。当前基于LLM的AI科学助理对相对狭窄的任务（例如支持文献评论）做出了相对较小的贡献。在近期情景中，它们在这些任务上变得更好，并具有更有影响力的能力，例如帮助产生强大的假设，或帮助预测实验的结果。但是，当前的系统仍然在人类科学家依靠这些任务的更深层次的创造力和推理上挣扎。正在进行努力提高这些AI功能，例如，将LLM与逻辑扣除引擎相结合，就像我们的字母和字母度计数器2示例一样，但需要进一步突破。对于那些在湿实验室，与人类参与者相互作用或漫长的过程（例如监测疾病进程）中，加速或自动化实验的能力也将更难。尽管再次，其中一些领域正在进行工作，例如新

重的实验室机器人和自动实验室。即使AI Systems的功能提高，最大的边际利益也将来自它们部署到具有相对优势的用例中，例如能够从大型数据集中快速提取信息的能力 - 并有助于解决真正的瓶颈，例如上述概述的五个机会，而不是自动化人类科学家已经做得很好的任务。随着AI使更便宜，更强大的科学，对科学和科学家的需求也将增长。例如，最近的突破已经导致了诸如蛋白质设计，材料科学和天气预报等领域的新创业公司。与其他部门不同，尽管过去提出了相反的说法，但对科学的未来需求实际上是无限的。新的进步始终在科学知识地图中打开了新的，不可预测的区域，AI也会做类似的区域。正如赫伯特·西蒙（Herbert Simon）所设想的那样，AI系统也将成为科学研究的对象，科学家将在评估和解释其科学能力以及开发新型的人类科学系统中发挥领导作用，并在开发新型的人类科学系统中发挥作用。参数B。我们对科学努力的成分感兴趣的成分以及对科学的良好成分以及在个人研究层面上都需要的成分 - 塑造它们。我们采访的专家通常将我们组织为玩具模型的几种成分，我们称之为科学生产功能的AI。该生产功能并不意味着详尽，规定或整洁的线性过程。这些成分对许多人来说都是直观的，但是我们的采访揭示了许多关于它们在实践中外观的课程，我们在下面分享了1。问题选择：追求雄心勃勃的AI形问题科学进步取决于能够识别一个重要的问题并询问有关如何解决问题的正确问题。Venkatesh Narayanamurti和Jeffrey Y. Tsao在探索科学突破的起源时记录了问题和答案之间的相互和递归关系的重要性，包括提出雄心勃勃的新问题的重要性。我们的科学团队首先要思考潜在的研究问题是否足够重大，足以证明时间和资源的大量投资。我们的首席执行官Demis Hassabis有一个心理模型来指导这项评估：将所有科学视为知识树。我们对根部 - 基本的“根节点问题”（例如蛋白质结构预测或量子化学反应）特别感兴趣，如果解决的话，可以解锁全新的研究和应用分支。为了评估AI是否合适且添加剂，我们寻找具有某些特征的问题，例如巨大的组合搜索空间，大量数据以及明确的目标功能，以基准对抗的性能。通常，问题原则上适合AI，但尚未到位，需要存储以备以后使用。AlphaFold的最初灵感之一是，Demis在很多年前就与一个朋友着迷于蛋白质折叠问题的朋友进行了对话。许多最近的突破也表明了这一重要科学问题和AI方法达到成熟点的融合在一起。例如，我们的融合工作得到了一种新颖的增强学习算法，称为最大后验政策优化，该算法刚刚发布。除了我们的合作伙伴EPFL刚刚开发的新快速准确模拟器外，这使团队能够克服数据贫困挑战。除了选择正确的问题外，重要的是要在适当的难度级别指定它。我们的受访者强调，对AI的有力问题陈述通常是一个可以中间结果的问题。如果您选择一个太难的问题，那么您将不会产生足够的信号来取得进展。正确的正确依赖于直觉和实验。2.评估：投资于可以提供强大的性能信号的评估方法，并得到社区科学家使用评估方法的认可，例如基准，指标和竞争，以评估AI模型的科学能力。做得很好，这些评估提供了一种跟踪进度，鼓励方法创新并激发研究人员对科学问题的兴趣的方法。通常，需要多种评估方法。例如，我们的天气预报团队始于基于一些关键变量（例如表面温度）的初始“进度度量”，它们曾经用于“爬山”，或逐渐改善模型的性能。当该模型达到一定水平的性能时，他们使用1300多个指标进行了更全面的评估，这些指标受到欧洲中范围天气预报中心的评估记分卡的启发。在过去的工作中，团队了解到，AI模型有时可以以不良的方式在这些指标上获得良好的分数。例如，“模糊”预测（例如预测较大的地理区域内的降雨量）的惩罚要少于“尖锐”预测 - 例如预测与实际位置略有不同的位置中的暴风雨 - 所谓的“双元素”问题。为了提供进一步的验证，团队评估了其模型对下游任务的有用性，例如预测旋风的轨迹的能力，并表征“大气河流”的强度 - 狭窄的集中水分带，可能导致洪水。科学评估方法最有影响力的AI通常是由社区驱动或认可的。黄金标准是对蛋白质结构预测竞争的关键评估。双年展的CASP竞争由约翰·穆尔特（John Moult）教授和克尔兹托夫·菲德利斯（Krzysztof Fidelis）教授于1994年成立，他挑战了研究小组，以测试其蛋白质结构预测模型的准确性，以针对真实的，未发行的实验蛋白质结构。它已成为一个独特的全球社区，也是研究进度的催化剂，尽管很难快速复制。社区买入的需求还为发布基准提供了一个论坛，以便研究人员可以使用，批评和改进它们，但

是，这也会产生一个风险，即基准将“泄漏”到AI模型的培训数据中，从而降低其跟踪进度的有用性。没有完美的解决方案，至少需要定期进行新的公共基准。科学家、AI实验室和政策制定者还应探索新的方法来评估AI模型的科学能力，例如建立新的第三方评估组织，竞争，并能够对科学家的AI模型能力进行更多的开放式探测。

3. 计算：跟踪计算使用方式如何发展并投资于专业技能多种政府评论已经认识到，计算对AI、科学和更广泛的经济的重要性越来越重要。随着我们在下面的进一步扩展，其能源消耗和温室气体排放也越来越重视。AI实验室和政策制定者应采取扎根的长期观点，以考虑计算需求如何在AI模型和用例中有所不同，潜在的乘数效应和效率提高，以及与不使用AI使用AI的科学进步的反事实方法相比。例如，某些最先进的AI模型（例如蛋白质设计中）相对较小。较大的模型，例如LLM，是训练的计算密集型，但通常需要更少的计算才能进行微调或进行推断，这可以为科学研究提供更有利的途径。一旦训练了LLM，就可以更容易地提高其效率，例如通过更好的数据策划，或将大型模型“提炼”到较小的模型中。与其他科学进步模型相比，还应评估AI计算需求。例如，AI天气预报模型是训练的计算密集型，但仍然比传统技术更有效率。这些细微差别凸显了AI实验室和政策制定者的需求，以经验跟踪计算使用，以了解其发展方式，并投射这些趋势对未来需求的意义。除了确保足够访问正确的芯片外，计算策略还应优先考虑管理访问并确保可靠性所需的关键基础架构和工程技能。在学术界和公共研究机构中，这通常不足。

4. 数据：将自上而下和自下而上的努力融合，以收集、策划、存储和访问计算的数据，数据可以看作是AI的关键基础架构，用于科学工作，需要开发、维护和更新随着时间的推移。讨论通常着重于确定决策者和从业者应创建的新数据集。这种自上而下的努力有角色。2012年，奥巴马政府启动了材料项目，以绘制已知和预测的材料，例如无机晶体，例如硅，它们在电池、太阳能电池板和计算机芯片中发现。我们最近的侏儒努力使用这些数据来预测220万个新型无机晶体，其中包括380,000个模拟在低温下稳定的，使其成为候选新材料的候选者。但是，通常很难提前预测哪些科学数据集将是最重要的，并且许多用于科学突破的AI依靠的数据依赖于更加有机化的小组，该数据依赖于小组的努力或小组的努力。例如，当时的Broad Institute的研究人员丹尼尔·麦克阿瑟（Daniel MacArthur）领导了我们的字母敏感工作随后启动的遗传变异的GNOMAD数据集。同样，数学证明助理和编程语言精益最初是由程序员莱昂纳多·德·穆拉（Leonardo de Moura）开发的。它不是数据集，但是现在许多AI实验室都使用它来帮助培训其AI数学模型，包括我们的字母隔板系统。诸如GNOMAD或LEAN之类的努力突出了如何在数据管道的各个阶段为个人提供更好的激励措施，如何补充自上而下的数据努力。例如，目前丢弃了战略湿实验室实验的一些数据，如果有稳定的资金，可以收集和存储。数据策划也可以更好地激励。我们的Alphafold模型接受了来自蛋白质数据库的数据的培训，该数据特别高质量，因为期刊需要将蛋白质结构沉积作为出版的先决条件，并且PDB的专业数据策展人为此数据制定了标准。在基因组学上，许多研究人员还必须将原始测序数据沉积在序列读取存档中，但不一致的标准意味着通常仍需要重新处理和组合单个数据集。由于限制性许可条件（例如在生物多样性中）或数据集未释放，例如从公共资助的融合实验中获得了数十年的数据，因此其他一些高质量的数据集完全没有使用。可能存在逻辑原因，例如缺乏时间、资金、放置数据的地方或为开发数据的研究人员临时禁运时期的需求。但是在汇总这些数据访问中，问题构成了一个关键的瓶颈，可以使用AI来提高科学进步。

5. 组织设计：在自下而上的创造力和自上而下的坐标之间取得正确的平衡，简单的启发式启发式是，学术界和行业倾向于在频谱的两端接近科学研究。学术界往往更自下而上，行业实验室往往更自上而下。实际上，长期以来，彼此之间有足够的空间，尤其是在最成功的实验室，例如贝尔实验室和施乐PARC的黄金时代，它们以其蓝天研究而闻名，并在DeepMind的建立中充当了灵感。最近，出现了新的科学研究机构，试图从这些异常值中学习。这些组织的目标、资助模式、纪律重点以及如何组织工作的情况有所不同，但是，他们总的来说，他们希望提供更

模式，无论焦点以及对于组织工作的情况有所不同。但是，他们想的来说，他们希望提供更多的高风险，高回报研究，更少的官僚主义和更好的科学家激励措施。许多人非常专注于应用AI，例如英国的高级研究和发明机构，ARC Institute以及旨在解决学术界太大的特定科学问题的越来越多的重点研究组织，例如对于行业而言，这些组织的盈利不足，例如对于扩大对AI Maths Research对AI Maths研究的精益助理的任务。这些新机构的核心渴望找到更好的自上而下的协调，并将重点与赋予科学家的自下而上的力量。对于某些组织而言，这意味着要专注于预先指定的里程碑的单个特定问题。对于其他人来说，这意味着向主要研究人员提供更多不受限制的资金。正确实现这种平衡对于吸引和留住研究领导者至关重要，如果要成功，他们也必须购买它 - Demis Hassabis将其视为成功协调大规模协调尖端研究的最大因素。在个人研究工作中，达到这种平衡也很重要。在Google DeepMind的情况下，通常在更无组织的“探索”阶段之间进行努力，在此阶段，团队搜索新想法和更快的“剥削”阶段，他们专注于工程和扩展性能。知道何时在这些模式之间切换以及如何相应地适应项目团队有一门艺术。

6.跨学科性：

作为一个团队的科学，资金被忽视的角色，并促进可竞争性的文化，许多最困难的科学问题都需要在领域之间的边界上取得进展。但是，当从业者聚集在一起时，例如在Covid -19期间，他们经常难以从多学科的过渡（他们每个人都保留自己的纪律角度）到真正的跨学科性，在那里他们集体发展共同的思想和方法。这一挑战反映了科学知识的不断增长以及诸如赠款之类的激励措施，这些挑战经常以其核心专业知识评估从业者。默认情况下，用于科学努力的人工智能通常是多学科的，但是要成功，他们需要成为真正的跨学科。一个起点是选择一个需要每种专业知识的问题，然后提供足够的时间和精力来培养团队围绕它的动态。例如，我们的ITHACA项目使用AI恢复和归因于损坏的古希腊铭文，这可以帮助从业者研究过去文明的思想，语言和历史。为了取得成功，共同领导的Yannis Assael项目必须发展对题词的理解 - 对古代铭文文本的研究。反过来，鉴于直觉对他们的工作的重要性，该项目的题生员不得不学习AI模型的工作原理。培养这些团队动力需要正确的激励措施。授权一支小巧的紧密联系团队专注于解决问题，而不是论文的作者，这是Alphafold 2突破的关键。这种类型的重点可以在行业实验室更容易实现，但再次强调了长期公共研究资金的重要性，而这种资金与出版压力相关。该图显示了有关雅典雅典座卫的恢复法令，日期为485/5公元前485/5。（CC BY-SA 3.0，Wikimedia）。要实现真正的跨学科性，组织还需要为可以帮助融合学科的个人创造角色和职业道路。在Google DeepMind，我们的研究工程师鼓励研究和工程之间的积极反馈循环，而我们的计划经理则帮助在研究工作中培养团队动态，并在其中建立联系。我们还优先考虑雇用喜欢在领域之间找到和桥接联系的个人，以及那些因在新领域迅速提高技能而激发的人。为了鼓励思想的交叉授粉，我们还鼓励科学家和工程师定期切换项目。最终，目标是建立一种鼓励好奇心，谦卑的文化，以及经济历史学家乔尔·莫基（Joel Mokyr）所说的“竞争性” - 所有背景的从业者都有能力在公开的谈判和讨论主题中表现和建设性地批评彼此的工作。

7.采用：

仔细考虑最佳的访问选项，并聚焦AI模型的不确定性许多AI用于科学模型（例如Alphafold或我们的天气预报工作），这是专门的，因为它们执行了少量的任务。但是，从理解疾病到改善捕鱼计划的所有方面，许多科学家都在使用它们，它们也是一般的。这种影响远非保证。疾病的细菌理论花了很长时间来扩散，而科学突破可以实现的下产品，例如新型抗生素，通常缺乏正确的市场激励措施。在决定如何发布我们的模型时，我们试图平衡具有商业目标和其他考虑因素（例如潜在安全风险）的科学家的广泛采用和验证的愿望。我们还创建了一个专门的影响加速器来推动突破性的采用，并鼓励可能不会发生的社会益处，包括通过与诸如被忽视疾病倡议的组织的合作伙伴关系以及全球抗生素研究与发展伙伴关系，这些抗生素研究与发展伙伴关系具有相似的授权。为了鼓励可以从新的模型或数据集中受益的科学家，开发人员需要使科学家尽可能容易地使用并将其集成到他们的工作流程中。考虑到这一点，对于Alphafold 2，我们开源了该代码，但也与EMBL-EBI合作开发了一个数据库，其中的科学家（包括计算技能和基础架构较少的科学家）可

以从预先存在的2亿个蛋白质结构的集合中进行搜索和下载。AlphaFold 3扩大了模型的功能，从而导致了潜在预测数量的组合爆炸。这创造了对新界面AlphaFold Server的需求，该服务器允许科学家按需创建结构。科学界还开发了自己的AlphaFold工具，例如Colabfold，证明了存在的需求多样性，以及在科学界培养计算技能以满足这些需求的价值。迄今为止，来自190多个国家/地区的200万用户访问了AlphaFold蛋白结构数据库，以查看超过700万个结构。Scientists还需要信任AI模型才能使用它。我们在下面的可靠性问题上扩展，但是一个有用的起点是主动指出科学家应如何使用模型及其不确定性。借助AlphaFold，在与科学家对话后，团队开发了不确定性指标，这些指标传达了该模型对给定蛋白质结构预测的“自信”，并得到了直观的可视化支持。我们还与Embl-Ebi合作开发了一个培训模块，该模块提供了有关如何最好地使用AlphaFold的指导，包括如何解释信心指标，并得到其他科学家如何使用它的实际例子。同样，我们的Med-Gemini系统最近在回答与健康相关的问题方面取得了最先进的表现。它使用不确定性引导的方法，通过为如何回答生成多个“推理链”来回答问题。然后，它使用这些初始答案之间的相对差异来计算答案的不确定。在不确定性较高的地方，它会调用Web搜索以集成最新的最新信息。Med-Gemini-3D能够生成用于CT扫描的报告，这是比标准X射线更复杂的成像形式。在此示例中，Med-Gemini-3D的报告正确地包括了原始放射科医生报告中错过的病理学（以绿色突出显示）。请注意，“基底”是Med-Gemini从培训数据中学到的“基础”的常见错误转录，尽管该报告的含义没有改变。

8. 合作伙伴关系：实现早期一致性和明确的价值交换AI进行科学努力，需要多种专业知识，这会在公共和私人组织之间建立强烈的伙伴关系 - 正式和非正式的伙伴关系。例如，我们的FunSearch方法开发了一个新的帽子套装问题的建筑，该结构著名的数学家Terence Tao曾经被描述为他最喜欢的开放问题。这是通过与威斯康星大学 - 麦迪逊分校的数学教授和著名的CAP SET专家合作的Jordan Ellenberg合作的。特别是，AI实验室通常需要科学家来帮助评估AI模型的输出。例如，我们的蛋白质设计团队与弗朗西斯·克里克研究所（Francis Crick Institute）的研究小组合作，运行湿实验室实验，以测试我们的AI设计的蛋白质是否与其靶标结合，并且是否具有所需的功能，例如防止SARS-COV-2感染细胞。鉴于行业实验室在提高AI功能方面的核心作用以及对丰富领域专业知识的需求，这些类型的公共私人合作伙伴关系对于推进AI的Science Frontier而言可能变得越来越重要，并且可能需要更多的投资，例如支持大学和公共研究机构的小伙伴团队的更多资金。建立伙伴关系很困难。在开始讨论时，重要的是要尽早保持整体目标并解决潜在的棘手问题，例如各方应对产出的权利应具有哪些权利，是否应该出版，是否应该开放模型或数据集，以及应采用哪种类型的许可。意见差异是自然的，并且通常反映了公共和私人组织面临的激励措施，而公共和私人组织面临的差异却大不相同，这取决于诸如研究的成熟或其商业潜力等因素。最成功的合作伙伴关系涉及一个明确的价值交换，该汇率借鉴了每个组织的优势。例如，来自190多个国家/地区的200万用户使用了AlphaFold蛋白质结构数据库。这需要密切合作，以将我们的AI模型与EMBL-EBI的生物疗法专业知识和科学网络相结合。

9. 安全与责任：使用评估来探索权衡取舍并激发新型评估方法的统计学家常常不同意AI模型对科学和更广泛的社会潜在利益和风险。进行伦理和安全评估可以帮助构建讨论，并使科学家能够决定是否以及如何开发给定的AI模型。一个起点是确定影响的最重要领域，并在正确的抽象级别指定这些域。越来越复杂的框架可以识别和分类不同的AI风险，例如实现错误和虚假信息。但是，这些框架很少考虑AI在同一领域中的潜在好处，例如改善获得高质量信息综合的访问或可能发生的权衡，例如，例如，如果您限制对AI模型的访问或限制其功能。评估还应阐明其时间尺度，任何影响的相对确定性以及AI的相对重要性或加法性。例如，那些担心AI和气候变化的人通常集中于培训大型AI模型所需的直接力量，而AI支持者通常专注于较不直接，不清晰的，但可能会从未来的AI应用程序带来的气候上的更大的下游益处。在进行评估时，AI从业人员还应避免对模型的功能过度索引，他们将更接近这些功能，并

更好地理解第三方实际使用它或受其影响的程度，这通常需要外部专家的投入才能做得很好。从业者还需要新的方法来更好地评估在科学中使用AI的潜在风险和好处。目前，许多AI安全评估依赖于指定模型不应输出的内容类型，并量化模型遵守此策略的程度。这些评估对于通过在科学中使用AI构成的某些风险很有用，例如产生不准确的内容。但是，对于其他风险，例如生物安全性，我们可以可靠地将某些类型的科学知识指定为危险的想法受到了挑战，因为科学知识的双重使用性质，但这也是因为这样的努力倾向于关注历史上造成危害的原因，例如过去暴发的病毒，而不是新颖的风险。一种更好的方法可能是评估AI模型的危险功能，或者AI模型为人类危险功能提供了提升的程度。在许多情况下，这些功能也将是双重使用，例如帮助设计或执行实验协议的能力。这些AI能力指出风险或机会的程度将取决于如何评估潜在威胁行为者以及如何访问模型。除了安全外，还要评估使用AI在科学中使用AI的其他风险，例如科学创造力或可靠性（我们在下面讨论），将需要全新的评估方法。鉴于难以研究和执行此类评估，在社区层面上追求它们是有意义的，而不是每个实验室追求孤立的努力。

C部分：科学家的风险散布文件，政府文件和调查会引用AI在科学中日益增长的使用风险。这些风险中的三个 - 科学的创造力，可靠性和理解 - 主要与科学的实践方式有关。对公平和环境的另外两个风险主要与科学的代表和影响更广泛的社会有关。通常将AI的使用仅作为对这些领域的风险，而诸如科学可靠性或环境之类的领域通常以稳定的，有点理想化的术语描绘，可以忽略它们面临的更广泛的挑战。我们认为，在科学中使用AI最终将受益于这五个领域中的每个领域，因为有机会减轻AI构成的风险，并利用AI来帮助解决这些领域的更广泛挑战，在某些情况下，深刻地。实现有益结果的不平等可能会更加困难，这是从劳动力的构成到基础研究的数据和科学创造力的数据，这是多个层次的AI和科学，这是高度主观的，因此个人可能会合理地不同意某个结果是否为积极。这些细微差别增加了科学家，政策制定者和其他人的价值，表达了他们对在科学中使用AI如何影响这5个领域的期望。

1. 创造力：AI会导致较小的新颖，违反直觉，突破吗？科学的创造力描述了创造有用的新事物。在实践中，科学家将新想法，方法或输出视为创造性的程度通常取决于更主观的因素，例如其感知的简单性，违反直觉或美丽。如今，科学的创造力受到科学劳动力的相对同质性的破坏，这使思想的多样性缩小了。研究人员“出版或灭亡”的压力还激励着“众群体”的出版物，即风险较小的话题，而不是那种经常构成创造性突破的深层工作或跨学科的概念的那种概念。这可以解释为什么导致领域转向新方向的破坏性科学思想的份额似乎正在下降，而随着科学的扩展，通常会超越通常的预期。一些科学家担心，通过破坏人类科学家的更直观，更直觉和偶然的方法，例如伽利略的假设，即地球在其轴上旋转，可能会加剧这些趋势。这可能以不同的方式发生。一个问题是，对AI模型进行了培训，以最大程度地减少其培训数据中的异常情况，而科学家通常会通过遵循有关令人困惑的数据点的直觉来扩大异常情况。其他人则担心AI系统经过培训以执行特定的任务，因此依靠它们将放弃更多偶然的突破，例如研究人员意外地发现解决方案的解决方案，以解决他们没有研究的问题。在社区层面上，有些人担心，如果科学家拥抱AI，它可能会导致产出的逐渐均匀化，例如，如果LLMs对不同科学家的查询产生类似的建议。或者，如果科学家过度关注学科和最适合AI的问题。维持对探索性研究和非AI研究的支持可能有助于减轻其中一些风险。科学家还可以量身定制他们的使用方式，以提高而不是损害自己的创造力，例如，通过微调LLM来提出更多个性化的研究思想，或帮助科学家更好地提出自己的想法，类似于我们早期的想法，类似于我们开发的AI教师，这些AI辅导者可以帮助学生更好地反映问题，而不是向问题提出回答，而不是向问题提出回答。AI还可以实现新型的科学创造力，这些创造力可能不太可能发生。AI创造力的一种类型是插值，AI系统在其训练数据中识别新颖的想法，尤其是在人类执行此操作的能力的情况下，例如使用AI来检测大型强子撞机实验中大量数据集中的异常情况。第二种是推断，其中AI模型推广到其训练数据以外的更新颖的解决方案，例如我们的AlphaGo系统提出的著名MOVE 37，震惊了人类GO专家，

或者新颖的数学证明和非显而易见的结构，我们的字母启动和AlphageSroof and AlphageMetrafe and Alphagemetry 2 System产生了该系统。第三类是发明，其中AI系统提出了一个全新的理论或科学系统，完全从其训练数据中删除，类似于一般相对论的原始发展或创建复数。AI系统目前并未证明这种创造力，但是新方法可能会解锁此功能，例如针对不同目标（例如新颖性和逆处之类）优化的多代理系统，或者经过训练以引起新科学问题的AI模型，以激发新的解决方案。

2. 可靠性：AI是否会减少科学的自我纠正？可靠性描述了科学家依靠彼此的发现的能力，并相信它们不是由于机会或错误所致。如今，一系列相互关联的挑战削弱了科学的可靠性，包括P黑客式和出版偏见，这可能会导致研究人员不足以报告负面影响；科学家如何执行常规科学任务缺乏标准化；错误，例如科学家如何使用统计方法；科学欺诈；同行审查过程（包括缺乏合格的同伴审阅者）所面临的挑战。一些科学家担心AI会加剧这些挑战，因为某些AI研究还具有不良实践，例如从业人员挑选了他们用于评估其模型表现的评估。AI模型，尤其是LLM，也容易出现“幻觉”输出，包括科学引用，这些输出是虚假或误导的。其他人则担心LLM可能会导致大量的低质量论文，类似于“造纸厂”耗散的论文。社区正在努力解决这些问题，包括良好的实践清单供研究人员遵守和不同类型的AI事实研究，例如培训AI模型，以将其输出以可信赖的来源为基础，或帮助验证其他AI模型的输出。科学家还可以使用AI来提高更广泛的研究基础的可靠性。例如，如果AI可以帮助自动化数据注释或实验设计的各个方面，则可以在这些领域提供急需的标准化。随着AI模型在将其输出置于引用方面变得更好，他们还可以帮助科学家和决策者对证据基础进行更多系统的评论，例如在气候变化中，诸如政府间气候变化的群体已经在努力与出版物中不可避免的上升相提并论。从业人员还可以使用AI来帮助检测错误或欺诈性的图像或误导科学主张，如AI图像分析工具的科学期刊最近的试验中所示。鉴于一些科学家已经使用LLM来帮助批评自己的论文，并帮助验证AI模型的输出，例如在定理中证明，AI可能会有可能帮助同行评审的各个方面。但是，鉴于同伴评论在诸如赠款批准之类的过程中所扮演的结果，也有合理的担忧，对机密性，AI系统检测真正的新工作的能力以及从科学家提供买入的能力。

3. 理解：AI会以更深入的科学理解为代价带来有用的预测吗？在最近的《自然调查》中，科学家们列举了对模式匹配的依赖，以更深入的理解为使用AI的最大风险。发现新的科学现象，例如超导性或开发有用的应用，例如药物，并不总是必要的。但是，大多数科学家将理解视为其主要目标之一，是人类知识的最深层形式。对AI破坏科学理解的担忧包括以下论点：现代深度学习方法是一种理论，并且不纳入或为他们预测的现象做出理论。科学家还担心AI模型是无法解释的，因为它们不是基于清晰的方程式和参数集。还担心，对AI模型的输出的任何解释都无法访问或对科学家有用。综上所述，AI模型可以提供有关蛋白质或天气结构的有用预测，但是他们是否能够帮助科学家了解为什么蛋白质以某种方式折叠，或者大气动态如何导致天气转移？担心用“低眉毛...”代替“真实的，理论科学”并不是什么新鲜事物，并且在过去的技术（例如蒙特卡洛方法）上进行了升级。合成生物学等合并工程和科学的领域也面临着将有用的应用优先于更深入的科学理解的指控。这些方法和技术导致了科学理解的收益，我们有信心AI也会，即使这些收益中的一些很难事先预测。首先，大多数AI模型不是理论上的，而是基于不同方式的先验知识，例如在其数据集和评估的构建中。一些AI模型也具有可解释的输出。例如，我们的FunSearch方法输出计算机代码，该代码还描述了它如何到达解决方案。研究人员还在研究解释性技术，这些技术可以阐明AI系统的工作方式，例如确定模型学习的“概念”的努力。这些解释性技术中有许多具有重要的局限性，但是它们已经使科学家从AI模型中提取新的科学假设。例如，转录因子是与DNA序列结合以激活或抑制附近基因表达的蛋白质。一项AI研究工作能够预测DNA序列中每个碱基对不同转录因子结合的相对贡献，并使用生物学家熟悉的概念来解释这一结果。一个更大的机会可能是根据AI系统的学习方式学习全新的概念。例如，我们的研究人员最近证明，我们的Alphazero系统学习了有关下棋的“超人”知识，包括非常规的动作和策略，并使用了只一个AI系统来提取这些概念并将其教给人类国际象棋专家，即使没有解

解释性技术，AI也会仅通过打开新的研究方向来改善科学理解，而这些方向否则会让人难以置信。例如，通过解锁产生大量合成蛋白质结构的能力，AlphaFold使科学家能够跨蛋白质结构进行搜索，而不是跨蛋白质序列。一组使用这种方法发现了Cas13蛋白家族的古老成员，该家族提供了编辑RNA的希望，包括帮助诊断和治疗疾病。这一发现还挑战了先前关于CAS13如何发展的假设。相反，修改AlphaFold模型体系结构以纳入更多先验知识的努力导致性能较差。这突出了准确性和可解释性之间可能发生的权衡，还强调了AI系统如何提高科学理解，而不是出于其不透明，而是由于这种不透明度，因为这种不透明度可能源于它们在可能对人类可能无法解释的高维空间中运作的的能力，而对于人类来说是必不可少的。

4. 公平：AI是否会使科学对边缘化群体的代表性和有用程度降低？在科学劳动力中，在他们研究的问题，开发的数据和模型以及所产生的益处和危害中，不平等在科学劳动力中明显可见。这些不平等是相关的，并且会随着时间的推移而复杂。例如，高收入城市中的少数实验室和个人在科学产出中占不变的份额。鉴定与疾病相关的遗传变异的研究在很大程度上取决于欧洲血统群体的数据，而被忽视的热带疾病不成比例地影响贫困国家的研究资金相对较少。在农业中，农作物创新集中在高收入国家中最常见的害虫上，然后不当地用于低收入国家的不同害虫，损害了收益率。尽管有积极的趋势，但妇女仅占科学家的33%，并且长期以来在临床试验中的人数不足，尤其是颜色的妇女。观察者担心AI在科学中的使用可能会加剧这些不平等。与许多其他科学学科相比，在性别，种族和领先实验室的位置方面，人工智能和计算机科学劳动力的代表性较低，因此AI的日益增长的使用可能会损害科学中更广泛的代表。作为一种数据驱动的技术，AI还冒着遗传和巩固科学数据集中发现的偏见的风险。也有机会使用AI来减少科学的不平等，尽管不能代替更多的系统性变化。如果通过低成本服务器或数据库提供AI模型，它们可能会使包括代表性不足的群体的科学家更容易，更便宜，以研究传统上被忽视的问题，类似于发布更多卫星数据的数据如何导致代表性不足的社区的更多研究。通过摄入更多的数据，AI模型还可以学习有关科学家研究的复杂系统的更多通用模式，使这些模型更加健壮，更容易偏向偏见。例如，由于它们的非代表性数据，鉴定与疾病相关的遗传变异的研究可能会引起混杂而不是因果变异。相反，在较大的蛋白质结构和遗传变异数据集上训练AI模型的一些早期尝试（包括跨物种数据）在预测疾病风险最大的个体方面的表现更好，并且人群群体之间的差异较少。然而，最终，提高公平将需要长期的努力，例如基因组学的H3Africa倡议和针对AI的深度学习Indaba倡议，旨在建立最缺乏的科学基础设施，社区和教育。

5. 环境：AI会伤害还是有助于实现Netzero的努力？鉴于他们渴望了解自然世界，许多科学家长期以来一直积极地保护环境，从而提供了有关气候变化的早期证据，以开发光伏细胞。近年来，越来越多的科学家对AI对环境的潜在影响表示担忧，并开发了试图量化这些影响的方法。大多数关注的侧重于训练和使用LLM对温室气体排放的潜在影响，以及相关问题，例如冷却数据中心所需的水。考虑这些效果的一种方法是生命周期方法，它捕获了直接和间接影响。直接效果包括构建和为AI模型训练和运行的数据中心和设备提供动力的排放。AI的所有直接排放都没有全面的估计。但是，2021年的估计表明，对许多大型AI模型进行了培训和部署的云和高度数据中心仅占全球排放的0.1-0.2%。随着LLM的大小不断增长，观察者警告说，这些数字可能会显着增加。但是，包括科学家在内的许多LLM的用户将能够以相对较低的计算成本来微调或使用其预测，而不是从头开始训练它们。也正在努力使LLM提高效率，并且数字技术的历史表明，可观的收益是可能的，这尤其是由于商业压力提供了更快，更便宜的AI模型。在某些情况下，AI模型的排放将低于其他方法。 For example, our internal analysis suggests that determining the structures of a small number (<10) of proteins experimentally uses roughly the same energy as a full training run of AlphaFold 2. These results need to be interpreted carefully, as AI simulations rely on, and inform, physical experiments, rather than substituting for them. But they also show how AI could enable a

larger amount of scientific activity at a lower average energy cost. Crucially, the direct effects of AI on emissions, whether positive or negative, will likely be minor compared to the indirect effects that AI-enabled applications have on emissions. Using AI in science opens up three major opportunities to reduce emissions. First, progress at the nexus between AI, maths and computer science could dramatically improve the efficiency of the Internet, from designing more efficient chips to finding more efficient algorithms for routine tasks. As a growing share of the economy moves online, this should help to offset emissions across these sectors. AI could accelerate the development and use of renewable energy, for example by designing new materials, such as for batteries or solar panels, by optimising how the grid operates and how it integrates renewables, and via more transformative but uncertain opportunities like fusion. Finally, the world is already getting warmer and AI could help to better prepare for extreme weather events. For example, our weather forecasting model recently correctly predicted, seven days in advance, that the deadly Hurricane Beryl would make ‘landfall’ in Texas. Non-AI models had originally predicted landfall in Mexico before correcting their prediction to Texas three days before it occurred.

Part D: The policy response

Given the importance of scientific progress to almost every major economic, environmental and security goal, it follows that science, and the potential for AI to accelerate it, should be a top priority for any government. What should a new AI for Science policy agenda look like? Policymakers can start by implementing the many good science and innovation policy ideas that already exist and which make even more sense in an era of AI-enabled science. For example, AI will improve the return on science research funding and so it provides a strong rationale to invest more in it and to trial new ideas to speed up and experiment with how this funding is allocated. On compute, governments could implement the idea set out in the UK’s Independent Review to empower a dedicated body to continually assess and advise governments on potential investments. To support AI for Science startups, policymakers can improve their spin-out policies and support well-run start-up incubators and fellowships. But ambitious new policies are also needed to capitalise on the AI for Science opportunity. We share four ideas below. They are intended to be widely applicable, though the precise details would need to be tailored to the specific context of a country, taking into account national priorities, unique strengths and the institutional landscape.

1. **Define the ‘Hilbert Problems’ for AI in Science** Scientific progress rests on picking the right problems. In 1900, the German mathematician David Hilbert published 23 unsolved problems that proved hugely influential for the subsequent direction of 20th century mathematics. As part of upcoming international events such as the AI Action Summit in Paris, policymakers, AI labs and science funders could launch a public call for scientists and technologists to identify the most important AI-shaped scientific problems, backed by a major new global fund to drive progress on them. Submissions should specify why the problem is important, why it is suited to modern AI systems, why it may be otherwise neglected, the data bottlenecks that exist, and how near-term technical progress could be evaluated. The top ideas could form the basis of new scientific competitions, where scientists compete to solve these problems with AI, supported by new datasets, evaluation methods and competitive benchmarks. These could build on the recent flurry of competitions that have emerged to evaluate the scientific capabilities of AI models, and include a new AI for Science Olympiad to attract exceptional young talent from across the world to the field. Beyond its direct impacts, the AI for Science ‘Hilbert Problems’ initiative could provide a welcome focal point for international scientific collaboration and funding, and inspire a new generation of interdisciplinary scientists to identify and pursue AI-shaped problems.
2. **Make the world readable to scientists** Most scientific data is uncollected, partial, uncurated or inaccessible, making it unavailable to train AI models. There is no single policy response to what is far from a uniform challenge. Policymakers and funders will need to blend a small number of top-down initiatives with support to scale up promising grassroots efforts. A new international network of AI for Science Data Champions could help to coordinate such efforts. The Data Champions initiative

Data Observatories should be set up to help address these goals. These Observatories could be given long-term backing and tasked with running rapid AI for Science ‘data stocktakes’, where expert teams map the state of data in priority disciplines and application areas. Stocktakes could identify existing datasets, such as the Sequence Read Archive, whose quality could be further improved, as well as untapped or underutilised datasets, such as the decades of experimental fusion data that is currently unavailable to scientists or leading biodiversity datasets that are subject to restrictive licensing conditions. The stocktakes could also include new ‘data wish lists’. For example, our internal analysis suggests that less than 7% of papers in key environmental research domains use AI. We recently funded Climate Change AI to identify datasets which, if available or improved, could remove some of the bottlenecks to higher AI use. To ensure this analysis leads to action, policymakers should designate and empower organisations to be accountable for addressing the results of the data stocktakes. The observatories could also scope the creation of new databases, including ensuring that adequate consideration is given to their long-term storage, maintenance, and incentives. This could include new databases to securely store the results of strategic wet lab experiments that are currently discarded, complemented by making the deposition of these experimental results a requirement for public research funding. Or digitising more public archives, following the example of a recent UK government and Natural History Museum collaboration to digitise their natural science collections, which includes more than 137 million items, from butterflies to legumes, across a 4.6 billion-year history. Policymakers can also empower scientists to use LLMs to create and improve their own datasets, by ensuring that publicly-funded research is open by default, where possible, building on recent examples from the UK, US and Japan, including mandates to release research via pre-print servers. Policymakers could seek to co-fund the most ambitious dataset initiatives with industry and philanthropy.

3. Teach AI as the next scientific instrument

Over the past half century, as the number of scientific technologies has grown, so has most scientists’ distance from them. Many technologies are the products of science, but an ever smaller share of scientists are trained in how to develop and use them effectively. The pressing near-term need is to fund and incentivise mass uptake of shorter, more tactical AI training programmes and fellowships, for existing scientists and research leaders. Policymakers can incentivise these efforts by setting a clear goal that every postgraduate science student should be able to access introductory courses on using AI in science, including on the most important tools in their domain, in the same way as basic statistics is often taught today. The type and depth of training needed will depend on an individual’s discipline and profile, and could range from basic introductory courses about how to reliably use LLMs for everyday research tasks, through to more advanced courses on how to fine-tune AI models on scientific data, as well as how to address more complex challenges, such as evaluating whether the data they used to test a model’s performance has intentionally or unintentionally ‘leaked’ into the data used to train it. These programmes could build on established examples such as the University of Cambridge’s Accelerate Programme that provides structured training in AI to PhD and postdoctoral researchers, or the short courses that The Carpentries offer on the programming, data, and computational skills needed to do research. Policymakers also need to move quickly to put in place longer-term programmes to ensure that the next generation of scientists has the skills they need. This means mainstreaming and deepening AI training and skills development in science education at all levels. Secondary school science students will need early exposure to the impact of AI while university students will need access to new types of interdisciplinary AI science degrees, such as the pan-African AI for Science Masters programme that we partnered with the African Institute for Mathematical Sciences to develop. Dedicated scholarships could also help. For example, the UK’s BIG Scholarships programme provides outstanding opportunities to high school students, with a focus on those from underrepresented groups who have excelled in mathematics and science. Observatories could also help to ensure that the next generation of scientists has the skills they need.

in international Science Symplads and Want to continue their study in leading science hubs but lack the funds to do so. 4. Build evidence and experiment with new ways of organising science Scientists’ use of AI is growing exponentially, but policymakers have little evidence about who is doing it best, how they are doing it, and the hurdles that are inhibiting others. This evidence gap is an impediment to identifying the best AI for Science policy ideas and targeting them effectively. Historically, answers to such questions often come from fields such as economics or innovation studies, but the results can take years to arrive. We are using citation data analysis, interviews, and community engagement to understand how scientists are using our AI models. Governments are also investing in these metascience capabilities to improve how they fund, share and evaluate science research. Building on this momentum, scientists could be tasked with a mission to rapidly assess foundational policy questions, including: where is the most impactful AI for Science research occurring and what types of organisations, talent, datasets, and evaluations are enabling it? To what extent are scientists using and fine-tuning LLMs vs more specialised AI models, and how are they accessing these models? To what extent is AI actually benefiting or harming scientific creativity, reliability, the environment, or other domains? How is AI affecting a scientist’s perception of their job and what skills, knowledge gaps, or other barriers are preventing their broader use of AI? Beyond informing robust policy responses, this evidence base will arm policymakers with the foresight they need to anticipate how AI will transform science and society, similar to the foresight they are developing for AI safety risks through the growing network of AI Safety Institutes. The evidence will also highlight opportunities to reimagine the incentives and institutions needed for science in the age of AI. In particular, scientists and policymakers have only explored a small fraction of the possible approaches to organising and executing science research. The rise of AI provides a welcome forcing function to experiment with new types of institutions, from those with more freedom to pursue high-risk, high-reward research, to Focused Research Organisations aimed at addressing specific bottlenecks. And from new types of interdisciplinary AI science institutes in priority domains such as climate or food security, to completely novel institutions that we are yet to imagine. Those who experiment faster will stand to benefit the most from a new golden age of discovery.

Acknowledgements

Thank you to Louisa Bartolo, Zoë Brammer and Nick Swanson for research support, and to the following individuals who shared insights with us through interviews and/or feedback on the draft. All views, and any mistakes, belong solely to the authors. Žiga Avsec, Nicklas Lundblad, John Jumper, Matt Clifford, Ben Southwood, Craig Donner, Joëlle Barral, Tom Zahavy, Been Kim, Sebastian Nowozin, Matt Clancy, Matej Balog, Jennifer Beroishi, Nitarshan Rajkumar, Brendan Tracey, Yannis Assael, Massimiliano Ciaramita, Michael Webb, Agnieszka Grabska-Barwinska, Alessandro Pau, Tom Lue, Agata Laydon, Anna Koivuniemi, Abhishek Nagaraj, Tom Westgarth, Guy Ward-Jackson, Harry Law, Arianna Manzini, Stefano Bianchini, Sameer Velankar, Ankur Vora, Sébastien Krier, Joel Z Leibo, Elisa Lai H. Wong, Ben Johnson, David Osimo, Andrea Huber, Dipanjan Das, Ekin Dogus Cubuk, Jacklynn Stott, Kelvin Guu, Kiran Vodrahalli, Sanil Jain, Trieu Trinh, Rebeca Santamaria-Fernandez, Remi Lam, Victor Martin, Neel Nanda, Nenad Tomasev, Obum Ekeke, Uchechi Okereke, Francesca Pietra, Rishabh Agarwal, Peter Battaglia, Anil Doshi, Yian Yin.

Errata & updates Note: In May 2025, we updated this essay to remove a reference to a now discredited study on the use of AI in material science by Aidan Toner-Rogers. Stuart Buck has a good write-up of the warning signs about this study, which we unfortunately missed. A good reminder to be appropriately skeptical with big claims!

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